Second Order Derivatives

Worked Solutions with TikZ Graphs

Wisest Maths — Year 1 A-Level, Differentiation (ref `d3`)

This document contains all 70 *Second Order Derivatives* questions, each with a fully worked solution and a TikZ diagram: the curve in blue with its **second derivative** $f''(x)$ in teal for the pure-computation and concavity questions (showing where the curve is concave up/down), and the curve with its **stationary points** marked and classified (max / min / point of inflection) for the second-derivative-test and optimisation questions.

Contents

Second Order Derivatives 01 (d3-001)

Difficulty: Foundation | Marks: 2

Find $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ when $y = x^4$.

Worked solution.

Step 1: Differentiate once to find $\frac{\mathrm{d}y}{\mathrm{d}x}$.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 4x^3$$

Apply the power rule: multiply by the index and reduce it by 1. The first derivative measures the gradient of the curve.

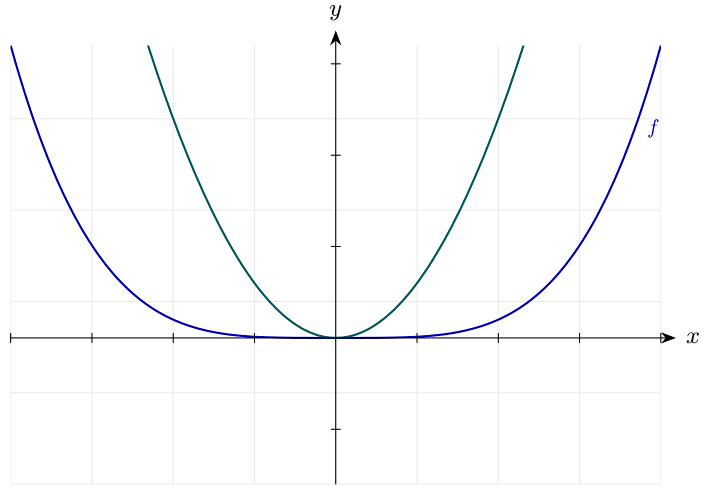
Step 2: Differentiate again to find $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12x^2$$

Differentiate $4x^3$: multiply 4 by 3 and reduce the index to 2. The second derivative is the rate of change of the gradient — it tells you how the gradient itself is changing as x increases.

Final answer: $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12x^2$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 02 `(d3-002)`

Difficulty: Foundation | Marks: 2

Find $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ when $y = 5x^3 - 2x$.

Worked solution.

Step 1: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 15x^2 - 2$$

Differentiate each term: $5x^3 \to 15x^2$ and $-2x \to -2$.

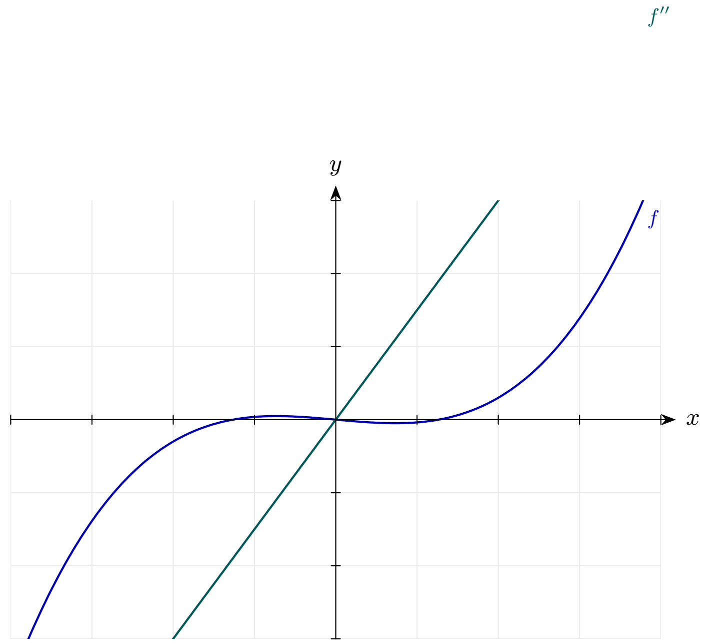
Step 2: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 30x$$

Differentiate $15x^2 - 2$: the constant -2 vanishes.

Final answer: $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 30x$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 03 `(d3-003)`

Difficulty: Foundation / *Marks: 2*

Find $f''(x)$ when $f(x) = 3x^4 - 6x^2 + 1$.

Worked solution.

Step 1: Find $f'(x)$.

$$f'(x) = 12x^3 - 12x$$

Differentiate each term.

Step 2: Find $f''(x)$.

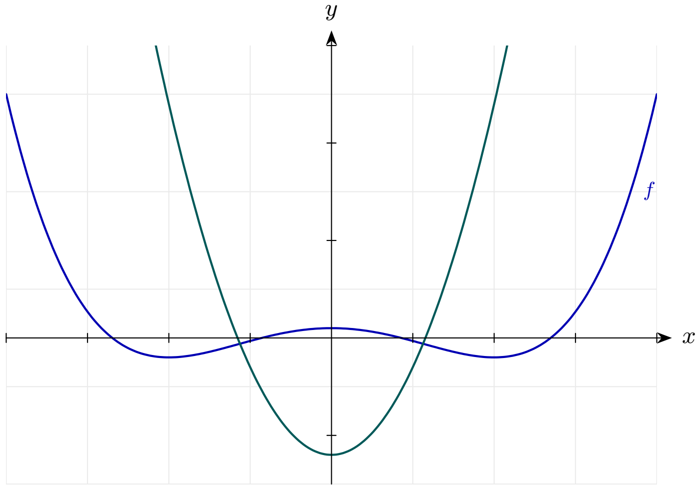
$$f''(x) = 36x^2 - 12$$

Differentiate $12x^3 - 12x$: the constant 1 in the original has already been lost.

Final answer: $f''(x) = 36x^2 - 12$

Diagram.

f''



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 04 `(d3-004)`

Difficulty: Foundation | Marks: 3

Find $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ when $y = 4x^5 - 3x^3 + 7x - 2$.

Worked solution.

Step 1: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 20x^4 - 9x^2 + 7$$

Differentiate each term; the constant -2 disappears.

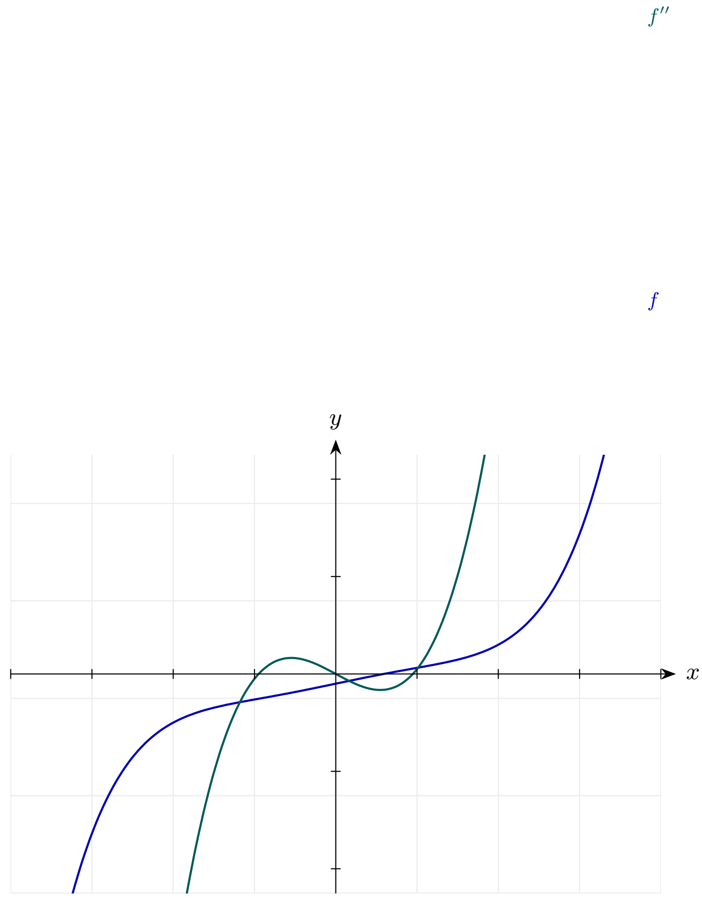
Step 2: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 80x^3 - 18x$$

Differentiate $20x^4 - 9x^2 + 7$; the constant 7 disappears.

Final answer: $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 80x^3 - 18x$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 05 `(d3-005)`

Difficulty: Foundation | *Marks: 3*

Find $f''(x)$ when $f(x) = x^{-2}$.

Worked solution.

Step 1: Find $f'(x)$ using the power rule.

$$f'(x) = -2x^{-3}$$

Multiply by the index -2 and reduce the index to -3.

Step 2: Find $f''(x)$.

$$f''(x) = (-2)(-3)x^{-4} = 6x^{-4}$$

Differentiate $-2x^{-3}$: two negatives multiply to give a positive.

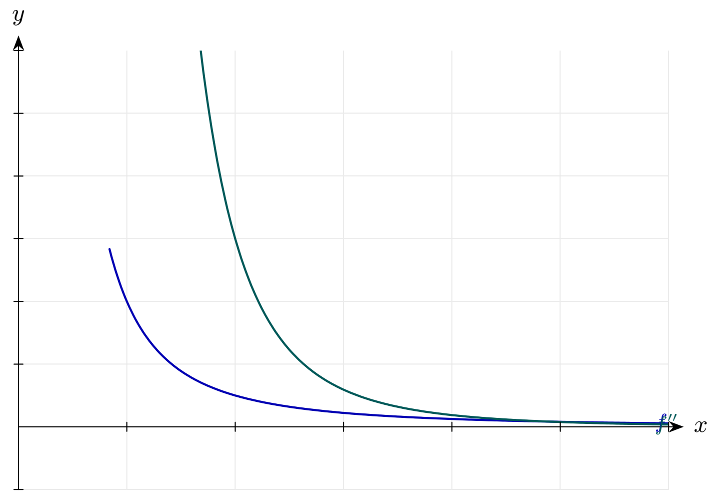
Step 3: Write in fraction form.

$$f''(x) = \frac{6}{x^4}$$

$x^{-4} = \dfrac{1}{x^4}$.

Final answer: $f''(x) = \dfrac{6}{x^4}$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 06 `(d3-006)`

Difficulty: Foundation | Marks: 3

Find $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ **when** $y = \sqrt{x}$**.**

Worked solution.

Step 1: Rewrite $\sqrt{x}$ *as* $x^{1/2}$.

$$y = x^{1/2}$$

Always convert surds to fractional powers before differentiating.

Step 2: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \tfrac{1}{2}x^{-1/2}$$

Power rule: multiply by $\frac{1}{2}$ and reduce index to $-\frac{1}{2}$.

Step 3: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \tfrac{1}{2} \cdot \left(-\tfrac{1}{2}\right) x^{-3/2} = -\tfrac{1}{4}x^{-3/2}$$

Differentiate $\frac{1}{2}x^{-1/2}$: reduce the index to $-\frac{3}{2}$.

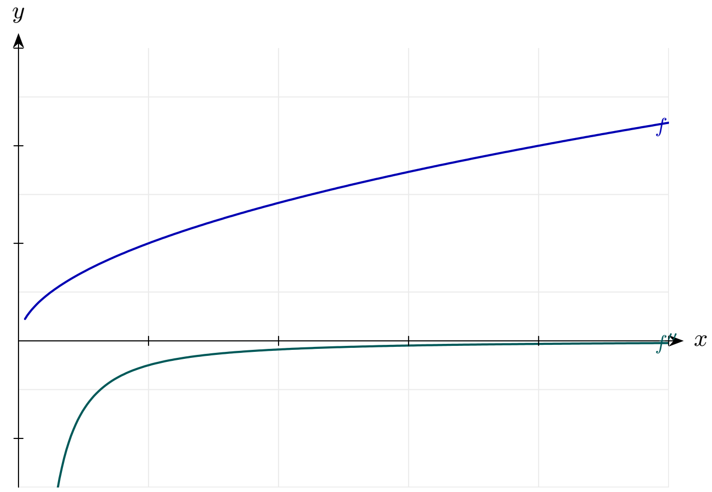
Step 4: Rewrite.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{1}{4x^{3/2}}$$

$x^{-3/2} = \dfrac{1}{x^{3/2}}$.

Final answer: $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\dfrac{1}{4x^{3/2}}$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 07 `(d3-007)`

Difficulty: Foundation / *Marks: 3*

Find $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ when $y = 6x^2 - \dfrac{1}{x}$.

Worked solution.

Step 1: Rewrite using index notation.

$$y = 6x^2 - x^{-1}$$

$\dfrac{1}{x} = x^{-1}$.

Step 2: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 12x + x^{-2}$$

Derivative of $6x^2$ is $12x$; derivative of $-x^{-1}$ is $+x^{-2}$.

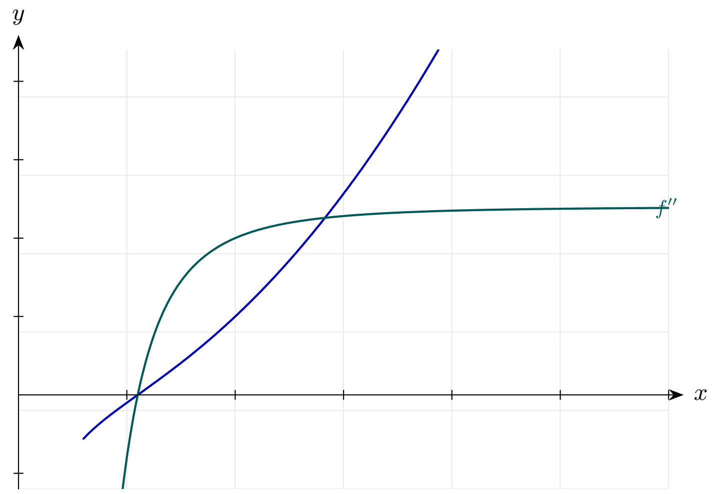
Step 3: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12 - 2x^{-3} = 12 - \frac{2}{x^3}$$

Differentiate $12x + x^{-2}$: derivative of x^{-2} is $-2x^{-3}$.

Final answer: $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12 - \dfrac{2}{x^3}$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 08 `(d3-008)`

Difficulty: Foundation | *Marks: 3*

Given $f(x) = 2x^3 - 9x^2 + 12x - 5$, find $f''(x)$ and hence find the value of x where $f''(x) = 0$.

Worked solution.

Step 1: Find $f'(x)$.

$$f'(x) = 6x^2 - 18x + 12$$

Differentiate term by term.

Step 2: Find $f''(x)$.

$$f''(x) = 12x - 18$$

Differentiate $6x^2 - 18x + 12$; the constant vanishes.

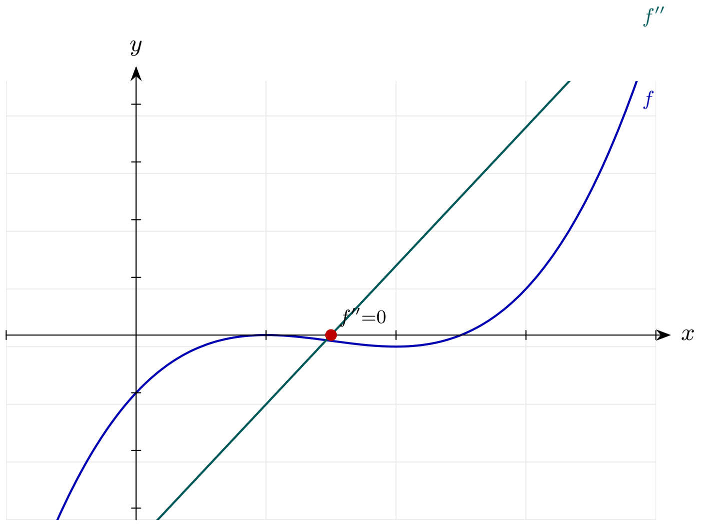
Step 3: Set $f''(x) = 0$.

$$12x - 18 = 0 \implies x = \frac{18}{12} = \frac{3}{2}$$

Solve the linear equation for x.

Final answer: $f''(x) = 12x - 18$, so $f''(x) = 0$ when $x = \dfrac{3}{2}$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 09 (d3-009)

Difficulty: Foundation | Marks: 3

Find $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ when $y = (x+2)(x-3)$.

Worked solution.

Step 1: Expand the brackets.

$$y = x^2 - x - 6$$

$(x+2)(x-3) = x^2 - 3x + 2x - 6$.

Step 2: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x - 1$$

Differentiate $x^2 - x - 6$.

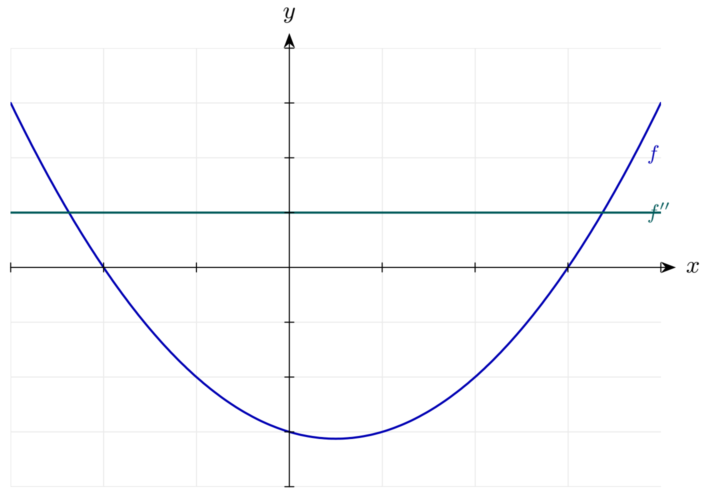
Step 3: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 2$$

Differentiating a linear function gives a constant.

Final answer: $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 2$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 10 (d3-010)

Difficulty: Foundation | Marks: 4

Find the second derivative of $y = \dfrac{x^4 - 2x^2}{x}$.

Worked solution.

Step 1: Simplify by dividing each term by x.

$$y = x^3 - 2x$$

$\dfrac{x^4}{x} = x^3$ and $\dfrac{-2x^2}{x} = -2x$.

Step 2: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 2$$

Power rule applied to each term.

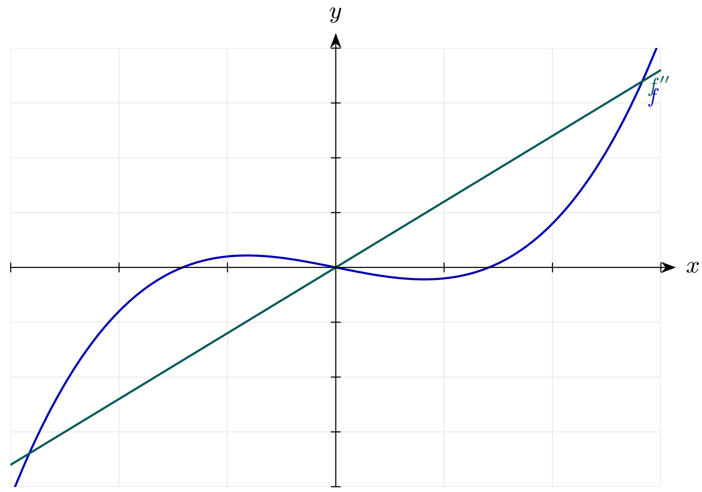
Step 3: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6x$$

Differentiate $3x^2 - 2$; the constant disappears.

Final answer: $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6x$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 11 `(d3-011)`

Difficulty: Foundation | *Marks: 3*

Find the value of $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ when $x = 2$, given that $y = x^3 - 5x$.

Worked solution.

Step 1: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 5$$

Power rule applied term by term.

Step 2: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6x$$

Differentiate $3x^2 - 5$.

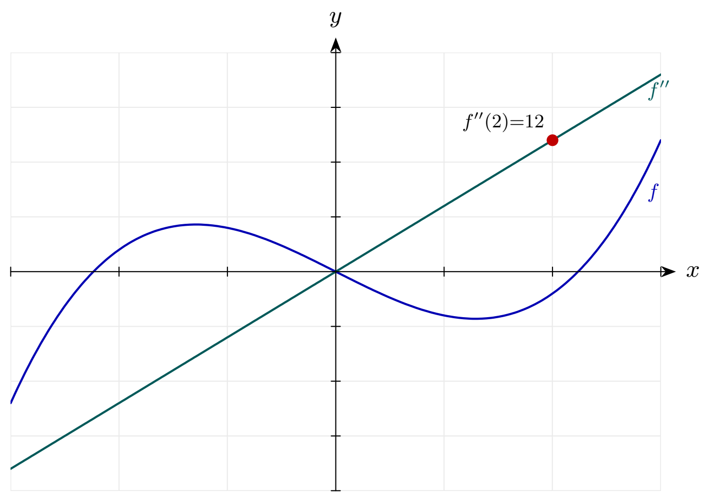
Step 3: Substitute $x = 2$.

$$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=2} = 6(2) = 12$$

Replace x with 2 in the second derivative expression.

Final answer: $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12$ when $x = 2$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 12 `(d3-012)`

Difficulty: Foundation | *Marks: 3*

Given $f(x) = x^4 - 8x^2$, find $f''(3)$.

Worked solution.

Step 1: Find $f'(x)$.

$$f'(x) = 4x^3 - 16x$$

Differentiate each term.

Step 2: Find $f''(x)$.

$$f''(x) = 12x^2 - 16$$

Differentiate $4x^3 - 16x$.

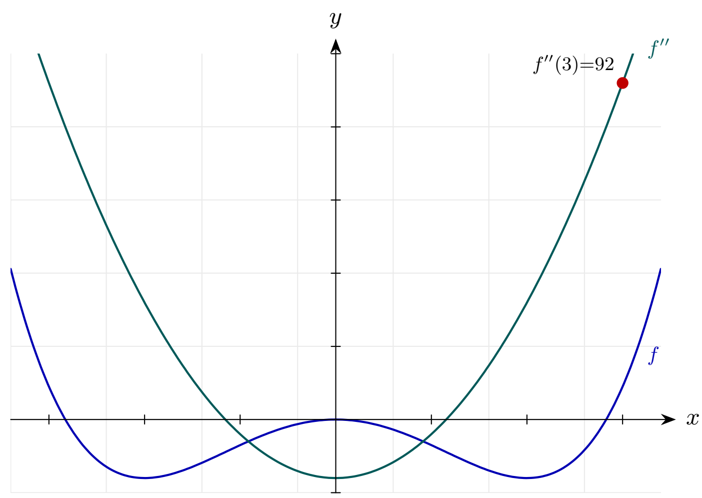
Step 3: Substitute $x = 3$.

$$f''(3) = 12(9) - 16 = 108 - 16 = 92$$

$3^2 = 9$.

Final answer: $f''(3) = 92$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 13 (d3-013)

Difficulty: Foundation | Marks: 3

Find $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ when $x = -1$, given $y = 2x^4 - x^3 + 5$.

Worked solution.

Step 1: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 8x^3 - 3x^2$$

Differentiate term by term; the constant 5 vanishes.

Step 2: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 24x^2 - 6x$$

Differentiate $8x^3 - 3x^2$.

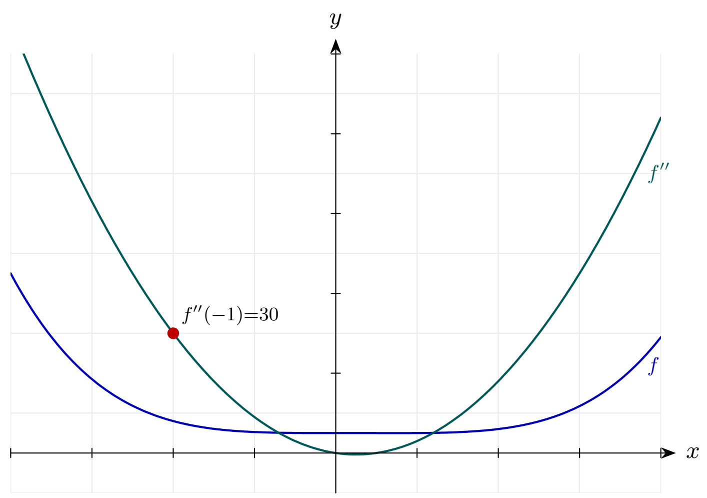
Step 3: Substitute $x = -1$.

$$24(-1)^2 - 6(-1) = 24 + 6 = 30$$

$(-1)^2 = 1$ and $-6 \times (-1) = +6$.

Final answer: $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 30$ when $x = -1$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 14 (d3-014)

Difficulty: Foundation | Marks: 6

For the curve $y = x^3 - 6x^2 + 9x + 1$, find the coordinates of the stationary points and use the second derivative test to determine their nature.

Worked solution.

Step 1: Find the first derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 12x + 9$$

Power rule applied to each term.

Step 2: Set $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ *and solve.*

$$3x^2 - 12x + 9 = 0 \implies x^2 - 4x + 3 = 0 \implies (x-1)(x-3) = 0$$

Divide by 3, then factorise.

Step 3: Find the y-coordinates.

$$x = 1 : y = 1 - 6 + 9 + 1 = 5; \quad x = 3 : y = 27 - 54 + 27 + 1 = 1$$

Substitute each x into the original equation.

Step 4: Find the second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6x - 12$$

Differentiate the first derivative.

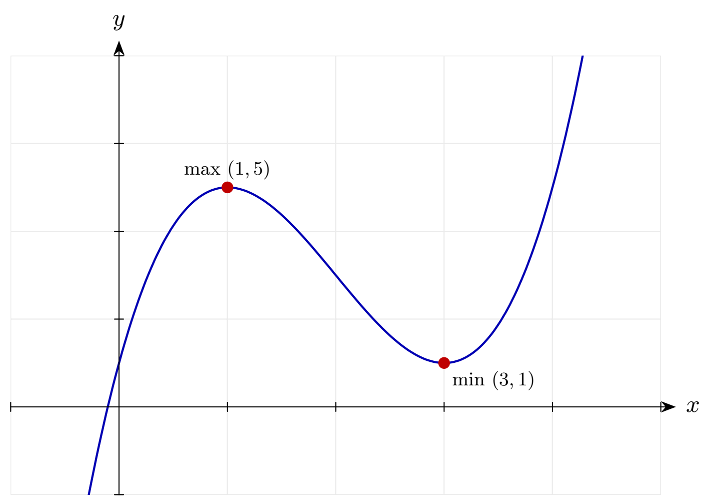
Step 5: Apply the second derivative test.

$$x = 1 : \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6(1) - 12 = -6 < 0 \Rightarrow \text{maximum} x = 3 : \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6(3) - 12 = 6 > 0 \Rightarrow \text{minimum}$$

The sign of $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ tells us about concavity: < 0 means concave down (local maximum), > 0 means concave up (local minimum). If it were $= 0$ the test would be inconclusive and we would need to inspect the gradient either side.

Final answer: Local maximum at $(1, 5)$; local minimum at $(3, 1)$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 15 (d3-015)

Difficulty: Foundation | Marks: 4

Find the stationary point of $y = x^2 - 8x + 3$ and use the second derivative to determine its nature.

Worked solution.

Step 1: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x - 8$$

Apply the power rule.

Step 2: Set equal to 0.

$$2x - 8 = 0 \implies x = 4$$

Solve the linear equation.

Step 3: Find the y-coordinate.

$$y = 16 - 32 + 3 = -13$$

Substitute $x = 4$ into the original curve.

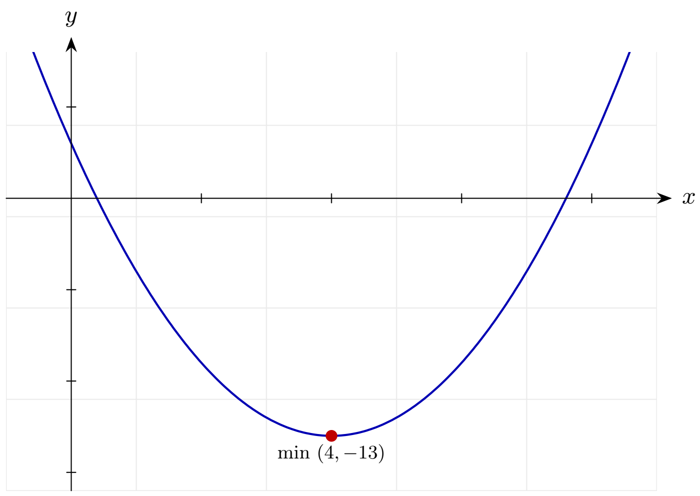
Step 4: Second derivative test.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 2 > 0$$

The second derivative is a constant $+2$. A positive value means the curve is concave up everywhere, so the stationary point must be a local minimum.

Final answer: Local minimum at $(4, -13)$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 16 (d3-016)

Difficulty: Foundation | Marks: 6

Find the stationary points of $y = 2x^3 - 3x^2 - 12x + 5$ and determine their nature using the second derivative test.

Worked solution.

Step 1: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 6x^2 - 6x - 12$$

Differentiate term by term.

Step 2: Solve $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$.

$$6x^2 - 6x - 12 = 0 \implies x^2 - x - 2 = 0 \implies (x-2)(x+1) = 0$$

Divide by 6, then factorise.

Step 3: Find y-coordinates at $x = 2$ and $x = -1$.

$$x = 2 : y = 16 - 12 - 24 + 5 = -15; \quad x = -1 : y = -2 - 3 + 12 + 5 = 12$$

Substitute each x back into y.

Step 4: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12x - 6$$

Differentiate the first derivative.

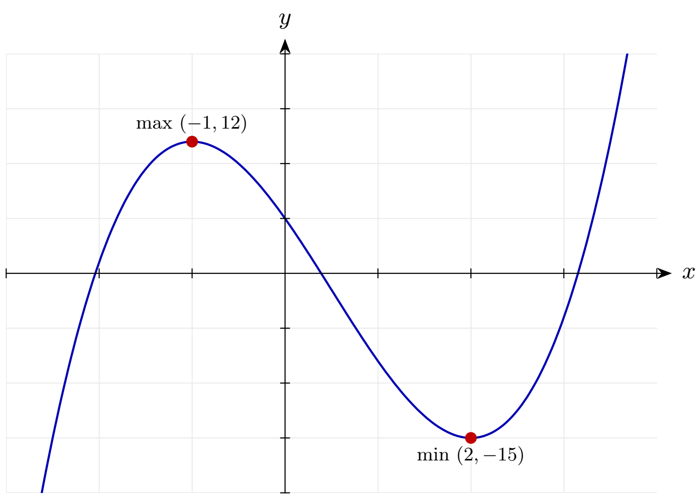
Step 5: Test each point.

$$x = 2 : 12(2) - 6 = 18 > 0 \Rightarrow \text{minimum} x = -1 : 12(-1) - 6 = -18 < 0 \Rightarrow \text{maximum}$$

A positive second derivative means concave up (local minimum); a negative one means concave down (local maximum). Watch the sign carefully when substituting negative x-values.

Final answer: Local minimum at $(2, -15)$; local maximum at $(-1, 12)$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 17 `(d3-017)`

Difficulty: Foundation | Marks: 4

Show that the curve $y = x^3$ has a stationary point of inflection at the origin.

Worked solution.

Step 1: Find the first derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2$$

Power rule.

Step 2: Stationary points: set $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$.

$$3x^2 = 0 \implies x = 0$$

There is exactly one stationary point at $x = 0$, which gives $y = 0$.

Step 3: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6x$$

Differentiate $3x^2$.

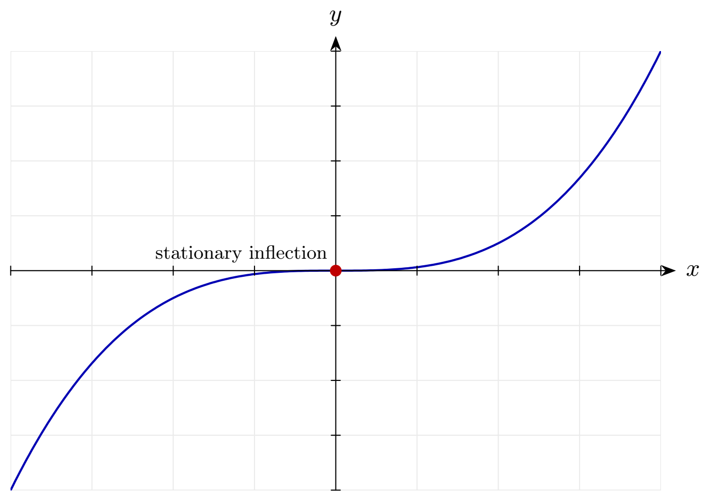
Step 4: Evaluate the second derivative at $x = 0$.

$$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=0} = 0$$

The second derivative is zero, so the test is inconclusive. Check behaviour either side: for $x < 0$, $\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 > 0$; for $x > 0$, $\frac{\mathrm{d}y}{\mathrm{d}x} > 0$. The gradient does not change sign, confirming a point of inflection.

Final answer: Stationary point of inflection at $(0, 0)$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 18 (d3-018)

Difficulty: Foundation | Marks: 4

Find the stationary point of $y = -x^2 + 6x - 4$ and determine its nature.

Worked solution.

Step 1: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -2x + 6$$

Differentiate $-x^2 + 6x - 4$.

Step 2: Set equal to 0.

$$-2x + 6 = 0 \implies x = 3$$

Solve for x.

Step 3: Find the y-coordinate.

$$y = -(9) + 18 - 4 = 5$$

Substitute $x = 3$ into the original.

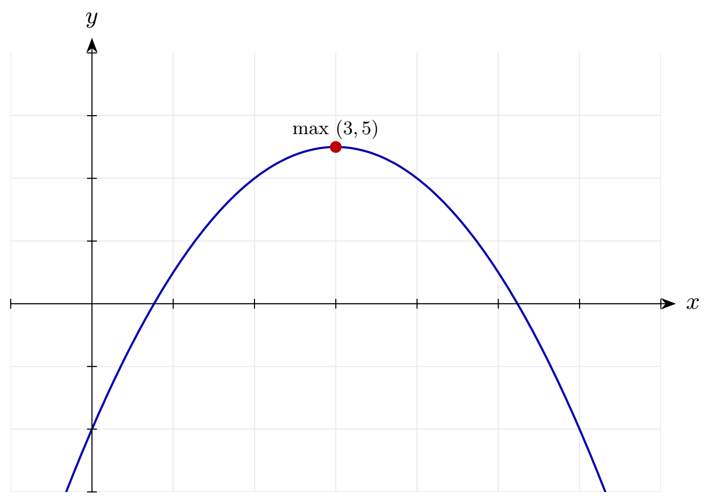
Step 4: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -2 < 0$$

The second derivative is a constant -2. A negative value means the curve is concave down everywhere, so the stationary point is a local maximum.

Final answer: Local maximum at $(3, 5)$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 19 (d3-019)

Difficulty: Foundation | Marks: 6

Find and classify the stationary points of $f(x) = x^4 - 4x^2$.

Worked solution.

Step 1: Find $f'(x)$.

$$f'(x) = 4x^3 - 8x$$

Power rule.

Step 2: Solve $f'(x) = 0$.

$$4x(x^2 - 2) = 0 \implies x = 0,\ x = \sqrt{2},\ x = -\sqrt{2}$$

Factorise: $4x^3 - 8x = 4x(x^2 - 2) = 0$.

Step 3: Find y-values.

$$f(0) = 0; \quad f(\sqrt{2}) = 4 - 8 = -4; \quad f(-\sqrt{2}) = -4$$

Substitute each x into $f(x) = x^4 - 4x^2$.

Step 4: Find $f''(x)$.

$$f''(x) = 12x^2 - 8$$

Differentiate $4x^3 - 8x$.

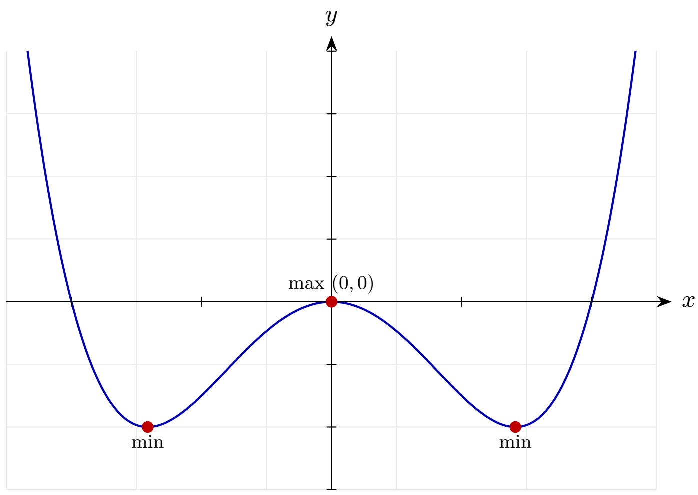
Step 5: Apply the test.

$$f''(0) = -8 < 0 \Rightarrow \text{maximum} f''(\pm\sqrt{2}) = 12(2) - 8 = 16 > 0 \Rightarrow \text{minima}$$

Substitute each x into $f''(x)$. Negative $\Rightarrow$ concave down (max); positive $\Rightarrow$ concave up (min). Note $(\pm\sqrt{2})^2 = 2$ — squaring kills the sign, so both $\pm\sqrt{2}$ give the same second-derivative value.

Final answer: Local maximum at $(0, 0)$; local minima at $(\sqrt{2}, -4)$ and $(-\sqrt{2}, -4)$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 20 (d3-020)

Difficulty: Foundation | Marks: 4

Find the second derivative of $y = 3x^{1/3}$ and write your answer without negative or fractional indices.

Worked solution.

Step 1: First derivative.

$$\frac{dy}{dx} = 3 \cdot \tfrac{1}{3}x^{-2/3} = x^{-2/3}$$

Multiply 3 by $\frac{1}{3}$ and reduce the index.

Step 2: Second derivative.

$$\frac{d^2y}{dx^2} = -\tfrac{2}{3}x^{-5/3}$$

Multiply 1 by $-\frac{2}{3}$ and reduce the index from $-\frac{2}{3}$ to $-\frac{5}{3}$.

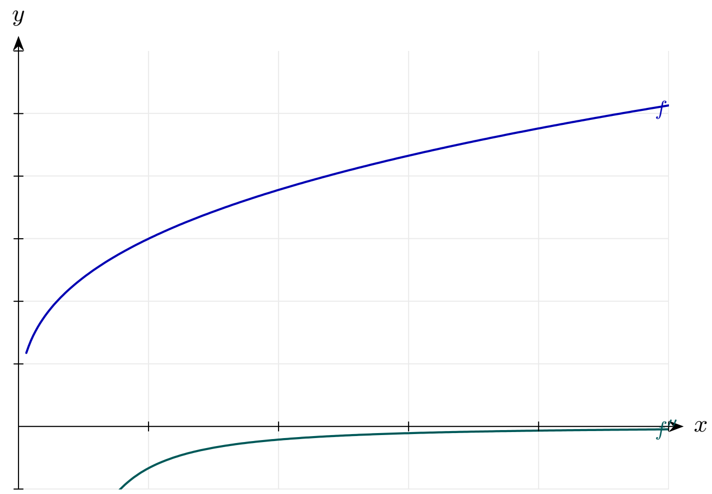
Step 3: Write without fractional or negative indices.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{2}{3x^{5/3}} = -\frac{2}{3\sqrt[3]{x^5}}$$

$x^{5/3} = \sqrt[3]{x^5}$.

Final answer: $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{2}{3\sqrt[3]{x^5}}$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 21 `(d3-021)`

Difficulty: Foundation | *Marks: 3*

Find the second derivative of $y = x(x^2 - 5)$.

Worked solution.

Step 1: Expand.

$$y = x^3 - 5x$$

Multiply x through the bracket.

Step 2: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 5$$

Power rule.

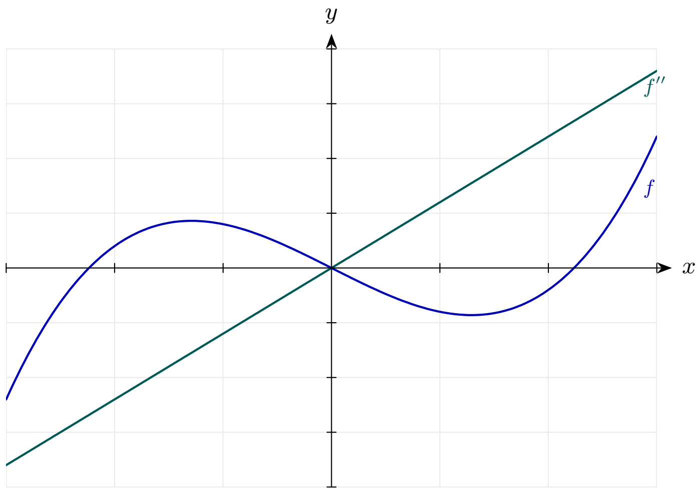
Step 3: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6x$$

Differentiate $3x^2 - 5$.

Final answer: $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6x$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 22 `(d3-022)`

Difficulty: Foundation | *Marks: 4*

Find the second derivative of $f(x) = \dfrac{x^5 + 3x^3}{x^2}$.

Worked solution.

Step 1: Simplify the fraction.

$$f(x) = x^3 + 3x$$

Divide each term in the numerator by x^2.

Step 2: Find $f'(x)$.

$$f'(x) = 3x^2 + 3$$

Differentiate term by term.

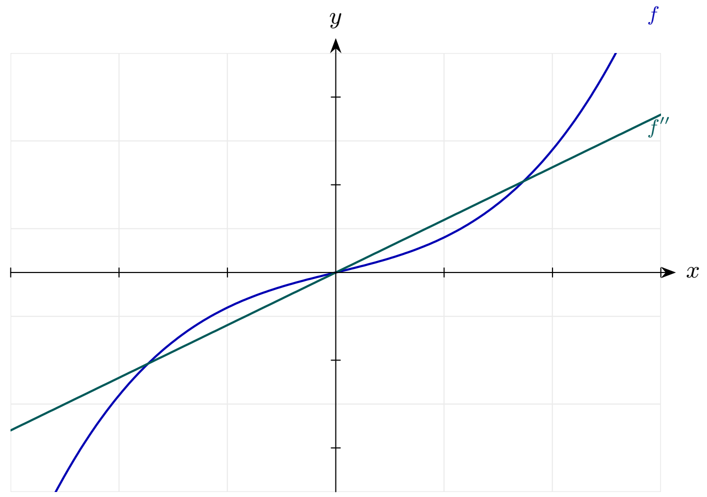
Step 3: Find $f''(x)$.

$$f''(x) = 6x$$

Differentiate $3x^2 + 3$; the constant vanishes.

Final answer: $f''(x) = 6x$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 23 `(d3-023)`

Difficulty: Foundation | *Marks: 6*

The curve C has equation $y = x^3 - 12x$. Find the coordinates of both stationary points. Use the second derivative to determine whether each is a maximum or minimum.

Worked solution.

Step 1: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 12$$

Power rule.

Step 2: Solve $3x^2 - 12 = 0$.

$$x^2 = 4 \implies x = \pm 2$$

Divide by 3 and take square roots.

Step 3: Find y-coordinates.

$$x = 2 : y = 8 - 24 = -16; \quad x = -2 : y = -8 + 24 = 16$$

Substitute into $y = x^3 - 12x$.

Step 4: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6x$$

Differentiate the first derivative.

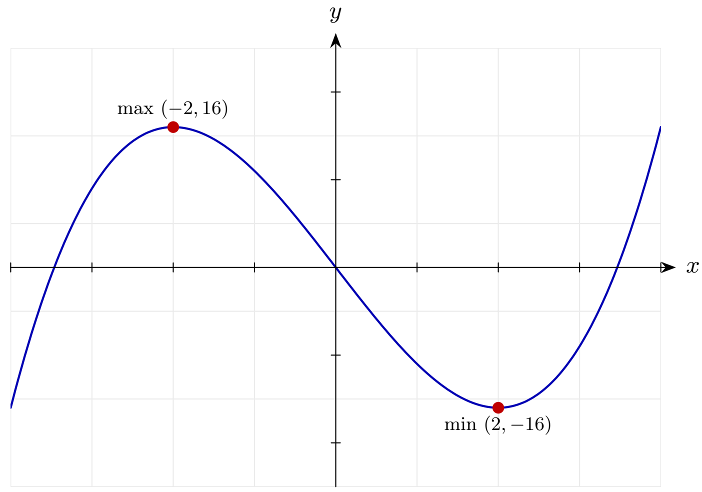
Step 5: Second derivative test.

$$x = 2 : 6(2) = 12 > 0 \Rightarrow \text{minimum} x = -2 : 6(-2) = -12 < 0 \Rightarrow \text{maximum}$$

Positive second derivative $\Rightarrow$ concave up (local minimum); negative $\Rightarrow$ concave down (local maximum).

Final answer: Local minimum at $(2, -16)$; local maximum at $(-2, 16)$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 24 `(d3-024)`

Difficulty: Foundation | *Marks: 4*

A function is defined as $f(x) = x^3 + px^2 + q$, where p and q are constants. Given that $f''(2) = 0$, find the value of p.

Worked solution.

Step 1: Find $f'(x)$.

$$f'(x) = 3x^2 + 2px$$

Differentiate; q is a constant so its derivative is 0.

Step 2: Find $f''(x)$.

$$f''(x) = 6x + 2p$$

Differentiate the first derivative; note q plays no role here.

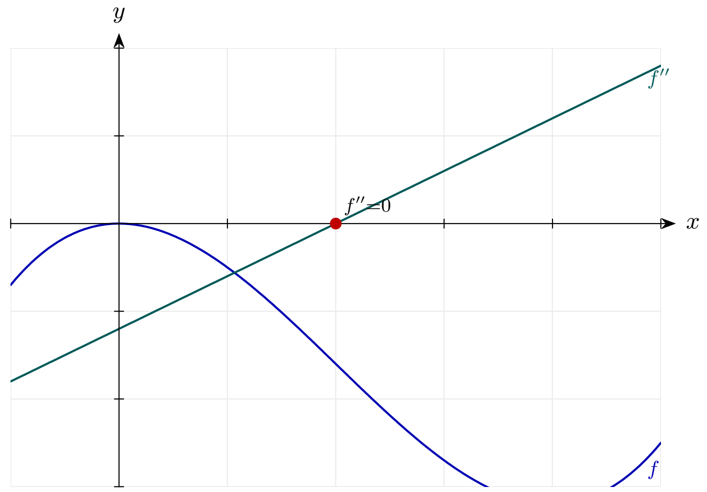
Step 3: Substitute $x = 2$ and set equal to 0.

$$f''(2) = 12 + 2p = 0 \implies p = -6$$

Solve the linear equation for p.

Final answer: $p = -6$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 25 `(d3-025)`

Difficulty: Foundation | *Marks: 5*

A curve has equation $y = ax^3 + bx$. Given that $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ at $x = 1$ and $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -6$ at $x = 1$, find the values of a and b.

Worked solution.

Step 1: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3ax^2 + b$$

Differentiate $y = ax^3 + bx$.

Step 2: Use $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ at $x = 1$.

$$3a + b = 0 \quad \cdots (1)$$

Substitute $x = 1$.

Step 3: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6ax$$

Differentiate the first derivative.

Step 4: Use $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -6$ at $x = 1$.

$$6a = -6 \implies a = -1 \quad \cdots (2)$$

Substitute $x = 1$.

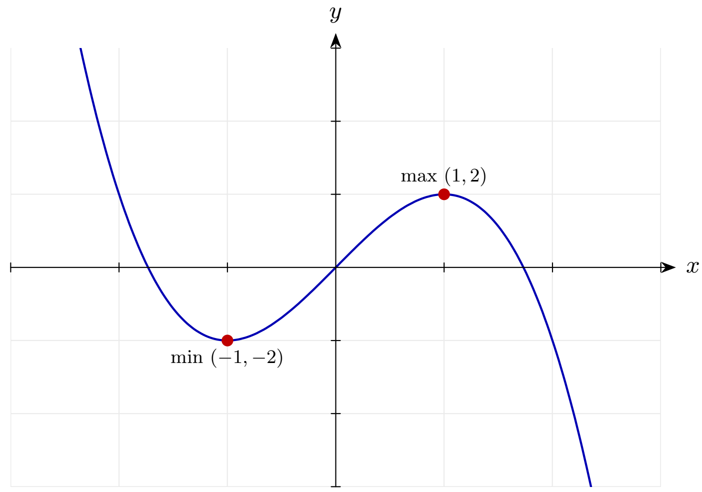
Step 5: Back-substitute $a = -1$ into equation (1).

$$3(-1) + b = 0 \implies b = 3$$

Solve for b.

Final answer: $a = -1,\ b = 3$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 26 `(d3-026)`

Difficulty: Foundation | Marks: 6

For $y = x^4 - 2x^3$, find both stationary points and determine their nature.

Worked solution.

Step 1: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 4x^3 - 6x^2$$

Power rule.

Step 2: Solve $4x^3 - 6x^2 = 0$.

$$2x^2(2x - 3) = 0 \implies x = 0 \text{ or } x = \tfrac{3}{2}$$

Factorise by taking out $2x^2$.

Step 3: Find y-values.

$$x = 0 : y = 0; \quad x = \tfrac{3}{2} : y = \tfrac{81}{16} - 2 \cdot \tfrac{27}{8} = \tfrac{81}{16} - \tfrac{54}{8} = \tfrac{81}{16} - \tfrac{108}{16} = -\tfrac{27}{16}$$

Substitute each x into $y = x^4 - 2x^3$.

Step 4: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12x^2 - 12x$$

Differentiate $4x^3 - 6x^2$.

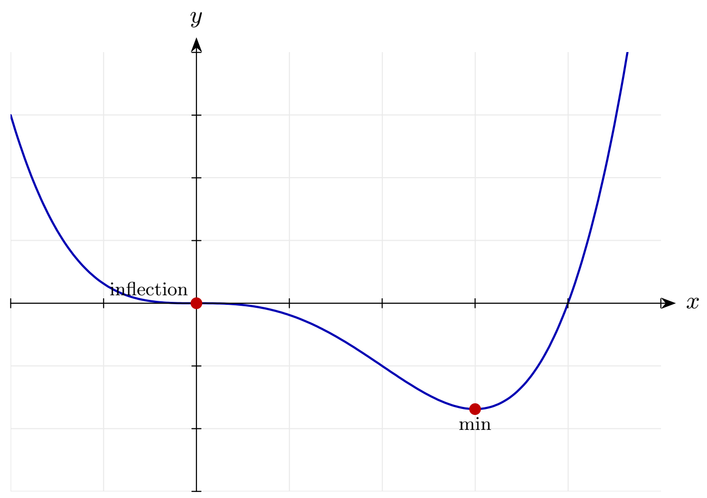
Step 5: Apply the test.

$x = 0 : 0 - 0 = 0 \Rightarrow$ inconclusive (point of inflection)$x = \frac{3}{2} : 12(\frac{9}{4}) - 12(\frac{3}{2}) = 27 - 18 = 9 > 0 \Rightarrow$ minimum

When $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$ the second-derivative test gives no information — you must fall back on checking the sign of $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ either side of the point. Here the gradient is negative on both sides of $x = 0$, so it does not change sign: this is a stationary point of inflection, not a turning point.

Final answer: Point of inflection at $(0,0)$; local minimum at $\left(\frac{3}{2}, -\frac{27}{16}\right)$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 27 `(d3-027)`

Difficulty: Foundation | Marks: 4

Verify that $y = 4x^3 - 3x^4$ has a local maximum at $x = 1$ by finding $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ and evaluating it there.

Worked solution.

Step 1: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 12x^2 - 12x^3$$

Differentiate term by term.

Step 2: Check $x = 1$ is indeed a stationary point.

$$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=1} = 12 - 12 = 0\checkmark$$

A stationary point requires $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$.

Step 3: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 24x - 36x^2$$

Differentiate $12x^2 - 12x^3$.

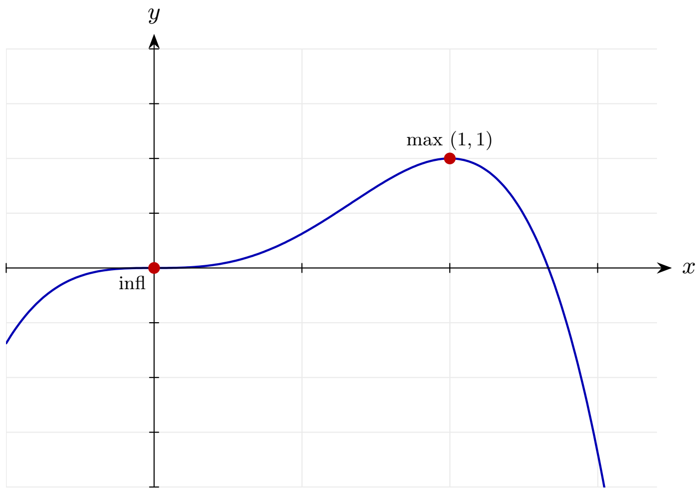
Step 4: Evaluate at $x = 1$.

$$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=1} = 24 - 36 = -12 < 0$$

A negative second derivative at the stationary point means the curve is concave down there, confirming a local maximum.

Final answer: $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = -12 < 0$ at $x = 1$, confirming a local maximum

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 28 `(d3-028)`

Difficulty: Foundation | *Marks: 4*

A function is given by $f(x) = 2x^3 - 3x^2 - 36x + 10$. (a) Find $f''(x)$. (b) Find the value(s) of x for which $f''(x) < 0$.

Worked solution.

Step 1: (a) Find $f'(x)$.

$$f'(x) = 6x^2 - 6x - 36$$

Differentiate term by term.

Step 2: Find $f''(x)$.

$$f''(x) = 12x - 6$$

Differentiate the first derivative.

Step 3: (b) Solve $f''(x) < 0$.

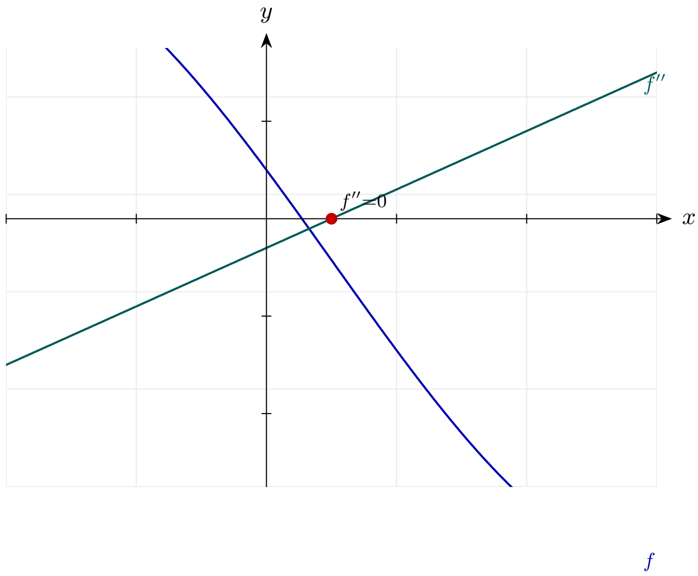
$$12x - 6 < 0 \implies 12x < 6 \implies x < \tfrac{1}{2}$$

Solve the linear inequality.

Final answer: (a) $f''(x) = 12x - 6$

(b) $x < \dfrac{1}{2}$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 29 `(d3-029)`

Difficulty: Foundation | Marks: 4

Find and classify the stationary point of $y = x^2 + 4x - 7$.

Worked solution.

Step 1: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x + 4$$

Differentiate term by term using the power rule.

Step 2: Set to 0.

$$2x + 4 = 0 \implies x = -2$$

Stationary points occur where the gradient is zero.

Step 3: Find y.

$$y = 4 - 8 - 7 = -11$$

Substitute $x = -2$ into the original equation to get the y-coordinate.

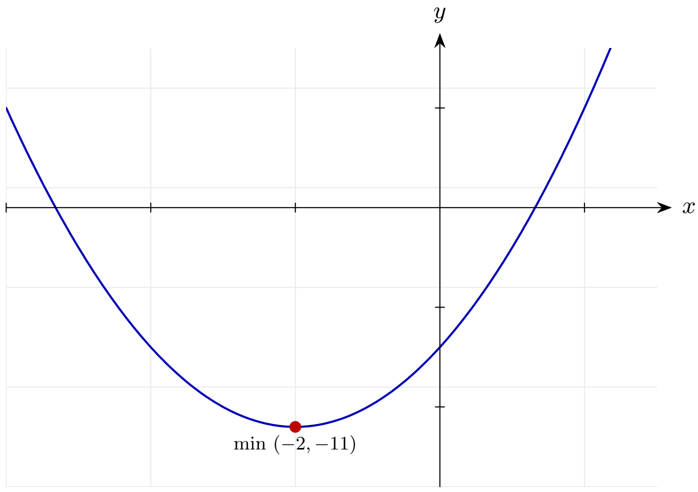
Step 4: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 2 > 0$$

The second derivative is a constant +2. A positive second derivative means the curve is concave up at the stationary point, so it is a local minimum.

Final answer: Local minimum at $(-2, -11)$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 30 (d3-030)

Difficulty: Foundation | Marks: 4

Given $y = 5x^2 - x^3$, find the range of values of x for which the curve is concave (i.e. $\frac{d^2y}{dx^2} < 0$).

Worked solution.

Step 1: First derivative.

$$\frac{dy}{dx} = 10x - 3x^2$$

Power rule.

Step 2: Second derivative.

$$\frac{d^2y}{dx^2} = 10 - 6x$$

Differentiate the first derivative.

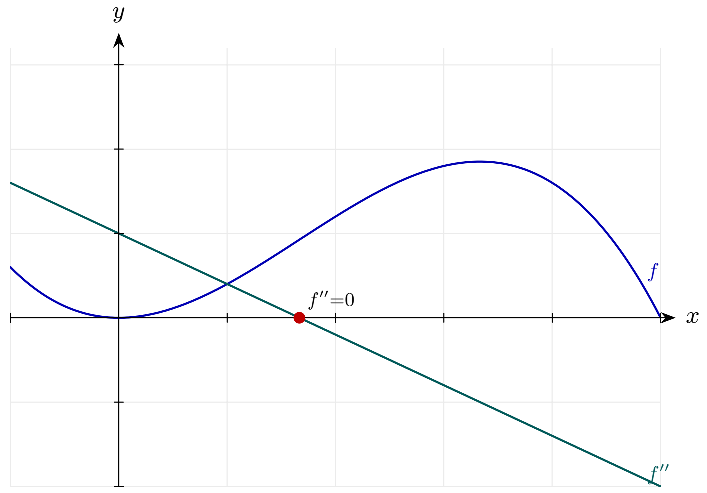
Step 3: Solve the inequality.

$$10 - 6x < 0 \implies 6x > 10 \implies x > \tfrac{5}{3}$$

A curve is concave (concave down) precisely where $\frac{d^2y}{dx^2} < 0$. Rearrange the inequality — no sign flip is needed because we divide by a positive number.

Final answer: $x > \frac{5}{3}$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 31 `(d3-031)`

Difficulty: Foundation | *Marks: 8*

A curve C has equation $y = x^3 - 3x^2 - 9x + 5$. (a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ and $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$. (b) Find the coordinates of the stationary points of C. (c) Determine the nature of each stationary point.

Worked solution.

Step 1: (a) Differentiate.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 6x - 9$$

Power rule applied term by term.

Step 2: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6x - 6$$

Differentiate the first derivative.

Step 3: (b) Set $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$.

$$3x^2 - 6x - 9 = 0 \implies x^2 - 2x - 3 = 0 \implies (x-3)(x+1) = 0$$

Divide by 3, then factorise.

Step 4: Find y-coordinates.

$$x = 3 : y = 27 - 27 - 27 + 5 = -22 x = -1 : y = -1 - 3 + 9 + 5 = 10$$

Substitute into the original equation.

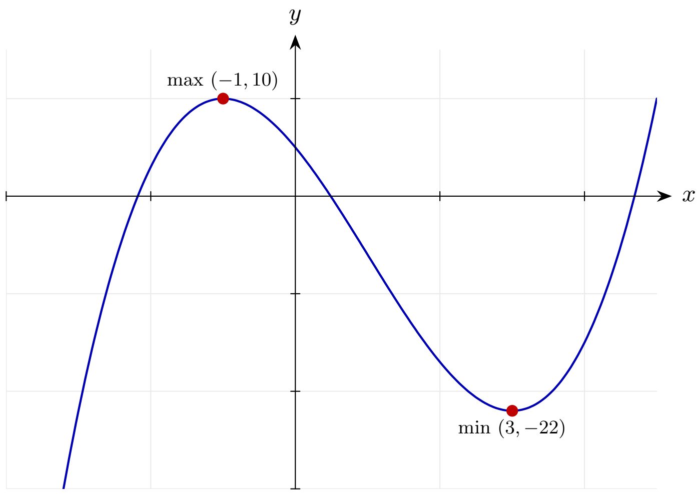
Step 5: (c) Apply second derivative test.

$$x = 3 : 6(3) - 6 = 12 > 0 \Rightarrow \text{minimum} x = -1 : 6(-1) - 6 = -12 < 0 \Rightarrow \text{maximum}$$

Substitute each x into $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6x - 6$. Positive $\Rightarrow$ concave up (minimum); negative $\Rightarrow$ concave down (maximum).

Final answer: (a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 6x - 9,\ \dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6x - 6$
(b) $(3, -22)$ and $(-1, 10)$
(c) Local minimum at $(3, -22)$; local maximum at $(-1, 10)$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 32 `(d3-032)`

Difficulty: Foundation | Marks: 8

A curve has equation $y = x^4 - 8x^2 + 3$. (a) Find all stationary points. (b) Use the second derivative to classify each one. (c) State the coordinates of the global minimum for $x \geq 0$.

Worked solution.

Step 1: (a) First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 4x^3 - 16x$$

Power rule.

Step 2: Solve $4x^3 - 16x = 0$.

$$4x(x^2 - 4) = 0 \implies x = 0,\ x = 2,\ x = -2$$

Factorise $4x(x-2)(x+2) = 0$.

Step 3: Find y-values.

$$y(0) = 3; \quad y(2) = 16 - 32 + 3 = -13; \quad y(-2) = -13$$

Substitute into $y = x^4 - 8x^2 + 3$.

Step 4: (b) Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12x^2 - 16$$

Differentiate the first derivative.

Step 5: Classify.

$$x = 0 : 0 - 16 = -16 < 0 \Rightarrow \text{maximum} x = \pm 2 : 48 - 16 = 32 > 0 \Rightarrow \text{minima}$$

Apply the second derivative test at each stationary point: < 0 means concave down (maximum), > 0 means concave up (minimum). Note $(\pm 2)^2 = 4$, so both endpoints share the same value.

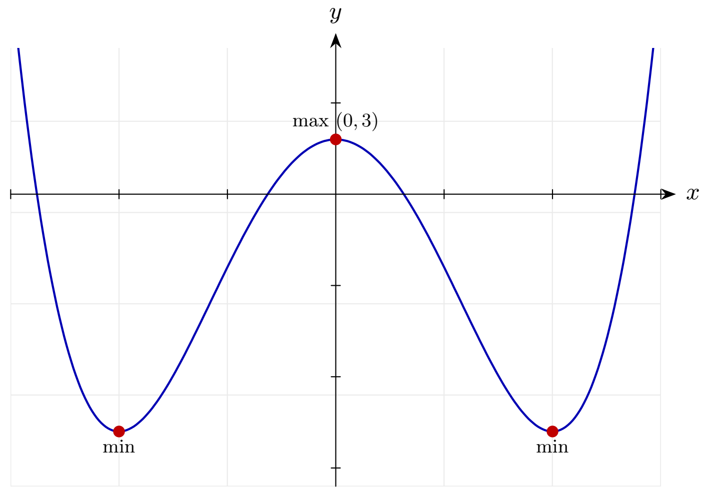
Step 6: (c) Global minimum for $x \geq 0$.

$$x = 2, \ y = -13$$

The only minimum for $x \geq 0$ occurs at $x = 2$.

Final answer: (a) Stationary points at $(0, 3)$, $(2, -13)$, $(-2, -13)$
(b) Maximum at $(0, 3)$; minima at $(2, -13)$ and $(-2, -13)$
(c) Global minimum for $x \geq 0$ at $(2, -13)$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 33 `(d3-033)`

Difficulty: Foundation | Marks: 6

The function $f(x) = \dfrac{k}{x^2} + 4x$, where k is a positive constant, has a stationary point at $x = 2$. (a) Show that $k = 16$. (b) Use the second derivative to confirm that this stationary point is a minimum.

Worked solution.

Step 1: Rewrite using index notation.

$$f(x) = kx^{-2} + 4x$$

$\dfrac{k}{x^2} = kx^{-2}$.

Step 2: (a) Find $f'(x)$.

$$f'(x) = -2kx^{-3} + 4$$

Power rule.

Step 3: Set $f'(2) = 0$ and solve for k.

$$-2k(2)^{-3} + 4 = 0 \implies -\tfrac{2k}{8} + 4 = 0 \implies \tfrac{k}{4} = 4 \implies k = 16$$

Substitute $x = 2$, using $(2)^{-3} = \frac{1}{8}$. A stationary point requires the first derivative to vanish, which fixes k.

Step 4: (b) Find $f''(x)$.

$$f''(x) = 6kx^{-4}$$

Differentiate $-2kx^{-3} + 4$: the power rule gives $-2k \cdot (-3)x^{-4} = 6kx^{-4}$; the constant 4 vanishes.

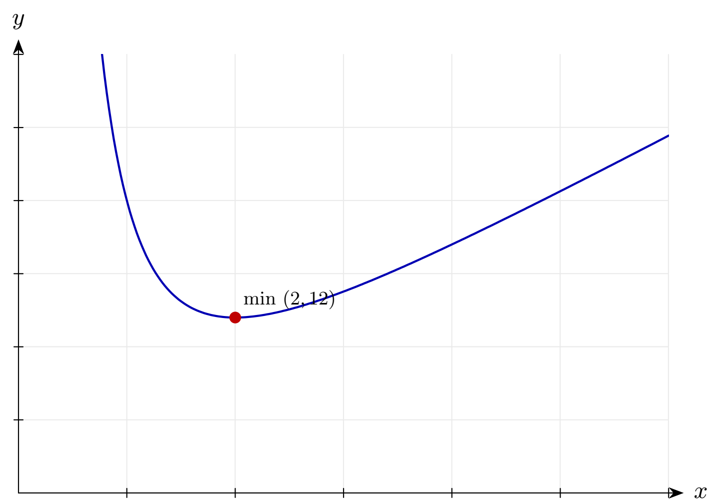
Step 5: Evaluate $f''(2)$ with $k = 16$.

$$f''(2) = 6(16)(2)^{-4} = 96 \cdot \tfrac{1}{16} = 6 > 0$$

The second derivative measures the rate of change of the gradient. Since $f''(2) > 0$, the curve is concave up at $x = 2$, so the stationary point is a local minimum.

Final answer: $k = 16$; $f''(2) = 6 > 0$, confirming a local minimum.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 34 (d3-034)

Difficulty: Foundation | Marks: 10

A curve C has equation $y = f(x)$, where $f(x) = 2x^3 + 3x^2 - 12x + 1$. (a) Find $f'(x)$ and $f''(x)$. (b) Find the coordinates of the stationary points of C. (c) Determine the nature of each stationary point using the second derivative. (d) Find the range of values of x for which $f''(x) > 0$.

Worked solution.

Step 1: (a) Find $f'(x)$.

$$f'(x) = 6x^2 + 6x - 12$$

Power rule applied term by term.

Step 2: Find $f''(x)$.

$$f''(x) = 12x + 6$$

Differentiate the first derivative.

Step 3: (b) Set $f'(x) = 0$.

$$6x^2 + 6x - 12 = 0 \implies x^2 + x - 2 = 0 \implies (x+2)(x-1) = 0$$

Divide by 6, then factorise.

Step 4: Find y-coordinates.

$$f(1) = 2 + 3 - 12 + 1 = -6 f(-2) = -16 + 12 + 24 + 1 = 21$$

Substitute $x = 1$ and $x = -2$ into $f(x)$.

Step 5: (c) Second derivative test.

$$f''(1) = 12 + 6 = 18 > 0 \Rightarrow \text{minimum} f''(-2) = -24 + 6 = -18 < 0 \Rightarrow \text{maximum}$$

Substitute each x into $f''(x) = 12x + 6$. Positive (concave up) gives a minimum; negative (concave down) gives a maximum.

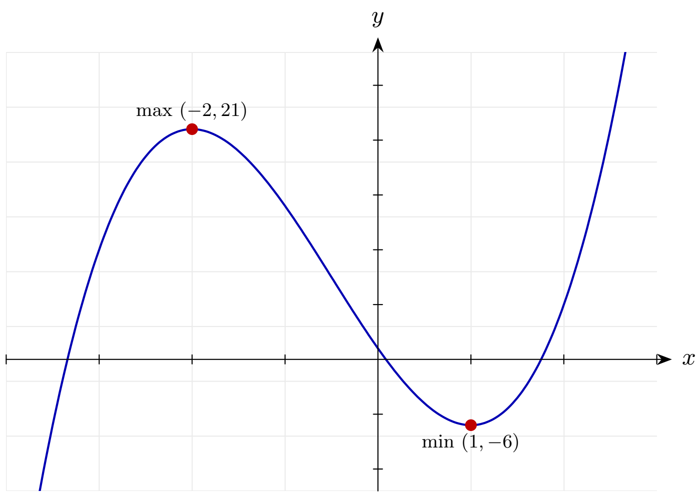
Step 6: (d) Solve $f''(x) > 0$.

$$12x + 6 > 0 \implies x > -\tfrac{1}{2}$$

Rearrange the inequality.

Final answer: (a) $f'(x) = 6x^2 + 6x - 12$, $f''(x) = 12x + 6$
(b) $(1, -6)$ and $(-2, 21)$
(c) Minimum at $(1, -6)$; maximum at $(-2, 21)$
(d) $x > -\dfrac{1}{2}$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 35 (d3-035)

Difficulty: Foundation | *Marks: 8*

A manufacturer models the profit P (in thousands of pounds) from producing x units (in hundreds) as $P(x) = -x^3 + 6x^2 + 15x$ for $0 \leq x \leq 8$. (a) Find $P'(x)$ and $P''(x)$. (b) Find the value of x that gives a stationary point in $(0, 8)$ and verify it is a maximum using the second derivative. (c) State the maximum profit.

Worked solution.

Step 1: (a) Find $P'(x)$.

$$P'(x) = -3x^2 + 12x + 15$$

Differentiate term by term.

Step 2: Find $P''(x)$.

$$P''(x) = -6x + 12$$

Differentiate the first derivative.

Step 3: (b) Set $P'(x) = 0$.

$$-3x^2 + 12x + 15 = 0 \implies x^2 - 4x - 5 = 0 \implies (x-5)(x+1) = 0$$

Divide by -3 (flip signs), then factorise.

Step 4: Select the stationary point in $(0, 8)$.

$$x = 5 \quad (x = -1 \text{ is outside the domain})$$

Only $x = 5$ lies in the interval $0 \leq x \leq 8$.

Step 5: Second derivative at $x = 5$.

$$P''(5) = -30 + 12 = -18 < 0$$

Negative second derivative confirms a local maximum.

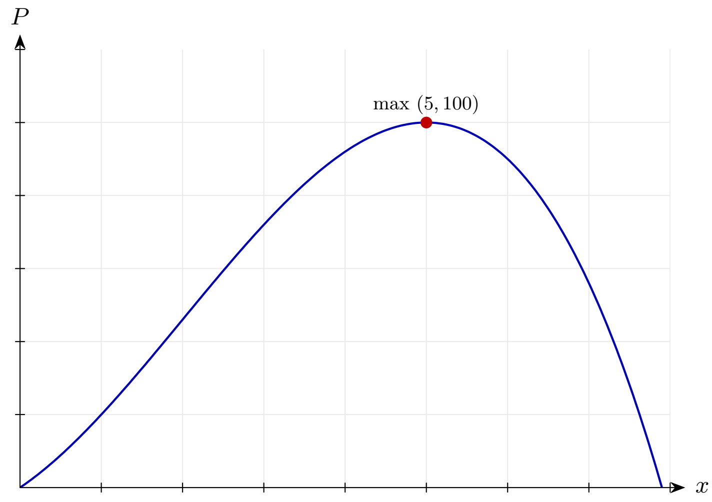
Step 6: (c) Maximum profit.

$$P(5) = -(125) + 6(25) + 15(5) = -125 + 150 + 75 = 100$$

Substitute $x = 5$ into $P(x)$.

Final answer: (a) $P'(x) = -3x^2 + 12x + 15$, $P''(x) = -6x + 12$
(b) Maximum at $x = 5$ ($P''(5) = -18 < 0$)
(c) Maximum profit = £100 000

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 36 (d3-036)

Difficulty: Standard | Marks: 3

Find $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ **when** $y = \dfrac{3}{x^2}$.

Worked solution.

Step 1: Rewrite with a negative index.

$$y = 3x^{-2}$$

Prepare for the power rule.

Step 2: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -6x^{-3}$$

Apply the power rule.

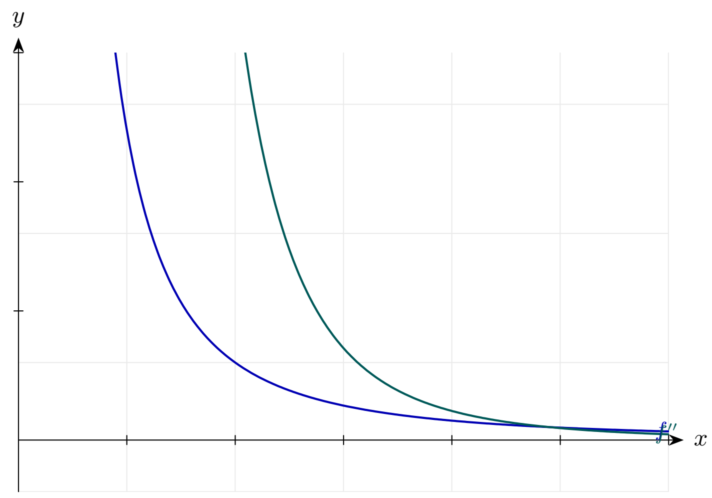
Step 3: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 18x^{-4} = \frac{18}{x^4}$$

Differentiate $-6x^{-3}$ and rewrite with a positive index.

Final answer: $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = \dfrac{18}{x^4}$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 37 `(d3-037)`

Difficulty: Standard | Marks: 3

Find $f''(x)$ when $f(x) = \dfrac{4}{\sqrt{x}}$.

Worked solution.

Step 1: Rewrite.

$$f(x) = 4x^{-1/2}$$

$\dfrac{1}{\sqrt{x}} = x^{-1/2}$.

Step 2: First derivative.

$$f'(x) = -2x^{-3/2}$$

Multiply by the index and reduce.

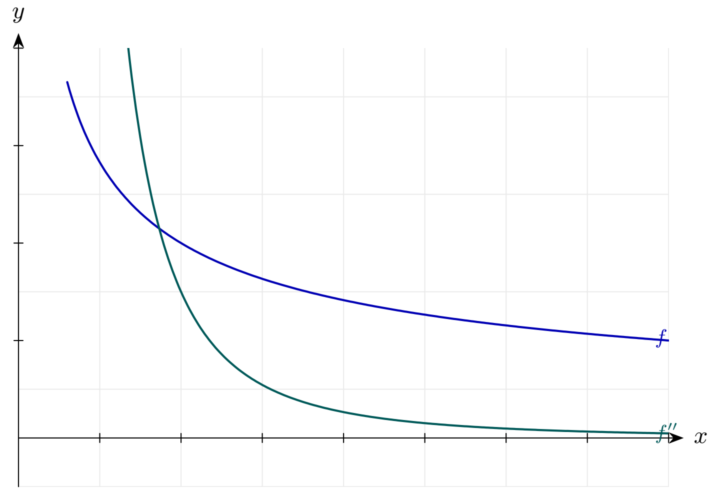
Step 3: Second derivative.

$$f''(x) = 3x^{-5/2} = \frac{3}{x^2\sqrt{x}}$$

Multiply -2 by $-\frac{3}{2}$ and reduce the index.

Final answer: $f''(x) = \dfrac{3}{x^2\sqrt{x}}$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 38 `(d3-038)`

Difficulty: Standard / Marks: 3

Find $\dfrac{d^2y}{dx^2}$ **when** $y = 2x^3 + \dfrac{5}{x}$**.**

Worked solution.

Step 1: Rewrite and differentiate once.

$$y = 2x^3 + 5x^{-1} \Rightarrow \frac{dy}{dx} = 6x^2 - 5x^{-2}$$

Apply the power rule term by term.

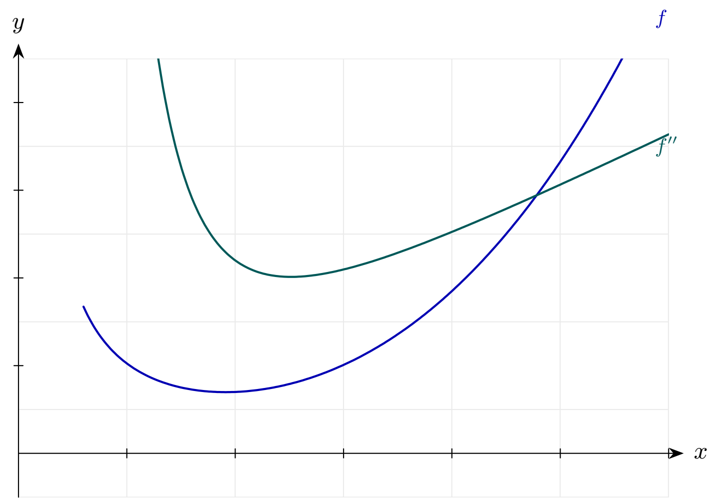
Step 2: Differentiate again.

$$\frac{d^2y}{dx^2} = 12x + 10x^{-3} = 12x + \frac{10}{x^3}$$

Apply the power rule to each term.

Final answer: $\dfrac{d^2y}{dx^2} = 12x + \dfrac{10}{x^3}$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 39 `(d3-039)`

Difficulty: Standard | *Marks: 3*

Given $y = x^{3/2} + 4x^{1/2}$**, find** $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$**.**

Worked solution.

Step 1: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \tfrac{3}{2}x^{1/2} + 2x^{-1/2}$$

Apply the power rule.

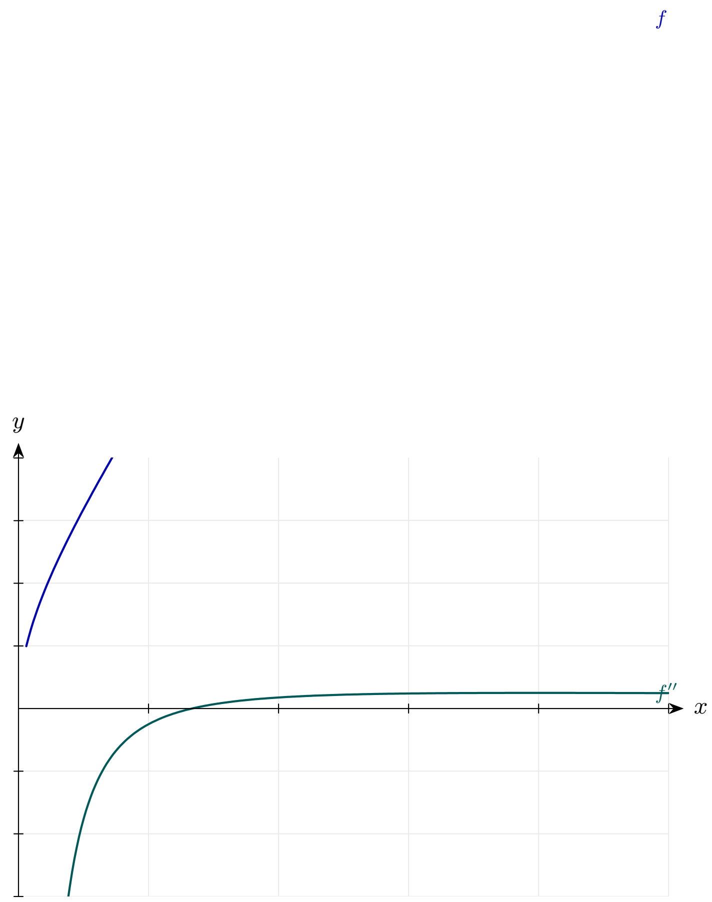
Step 2: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \tfrac{3}{4}x^{-1/2} - x^{-3/2} = \frac{3}{4\sqrt{x}} - \frac{1}{x\sqrt{x}}$$

Differentiate each term and simplify.

Final answer: $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = \dfrac{3}{4\sqrt{x}} - \dfrac{1}{x\sqrt{x}}$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 40 `(d3-040)`

Difficulty: Standard | *Marks: 4*

Find the second derivative of $f(x) = \dfrac{x^3 - 2x}{x^2}$ for $x \neq 0$.

Worked solution.

Step 1: Simplify first.

$$f(x) = \frac{x^3}{x^2} - \frac{2x}{x^2} = x - 2x^{-1}$$

Divide each numerator term by x^2.

Step 2: First derivative.

$$f'(x) = 1 + 2x^{-2}$$

Apply the power rule.

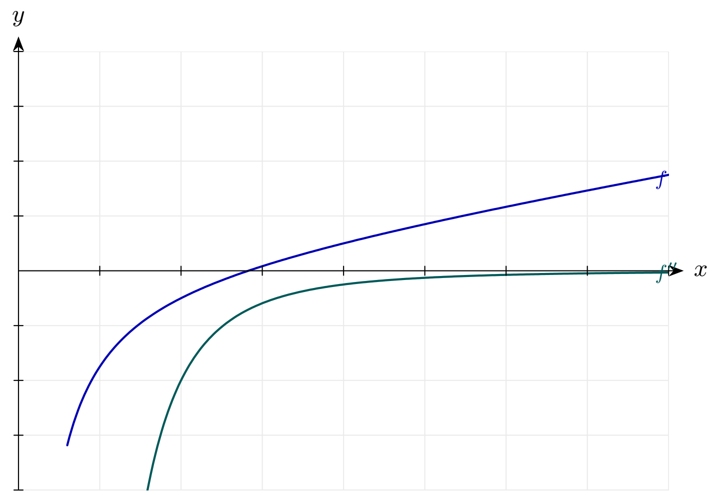
Step 3: Second derivative.

$$f''(x) = -4x^{-3} = -\frac{4}{x^3}$$

Differentiate once more.

Final answer: $f''(x) = -\dfrac{4}{x^3}$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 41 `(d3-041)`

Difficulty: Standard | *Marks: 3*

Find $\dfrac{d^2y}{dx^2}$ when $y = (x+3)^2$.

Worked solution.

Step 1: Expand.

$$y = x^2 + 6x + 9$$

$(a+b)^2 = a^2 + 2ab + b^2$.

Step 2: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x + 6$$

Apply the power rule.

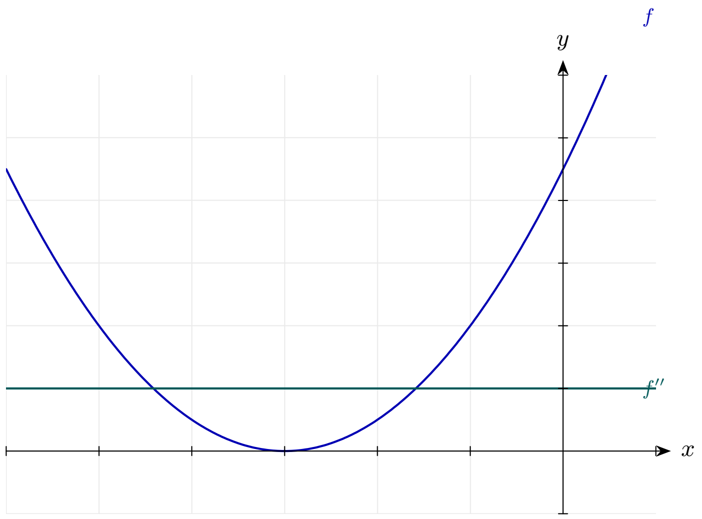
Step 3: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 2$$

Constant.

Final answer: $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 2$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 42 (d3-042)

Difficulty: Standard | Marks: 4

Find the second derivative of $y = \sqrt{x}(x-3)$.

Worked solution.

Step 1: Expand using indices.

$$y = x^{3/2} - 3x^{1/2}$$

$\sqrt{x} \cdot x = x^{3/2}$.

Step 2: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \tfrac{3}{2}x^{1/2} - \tfrac{3}{2}x^{-1/2}$$

Apply the power rule.

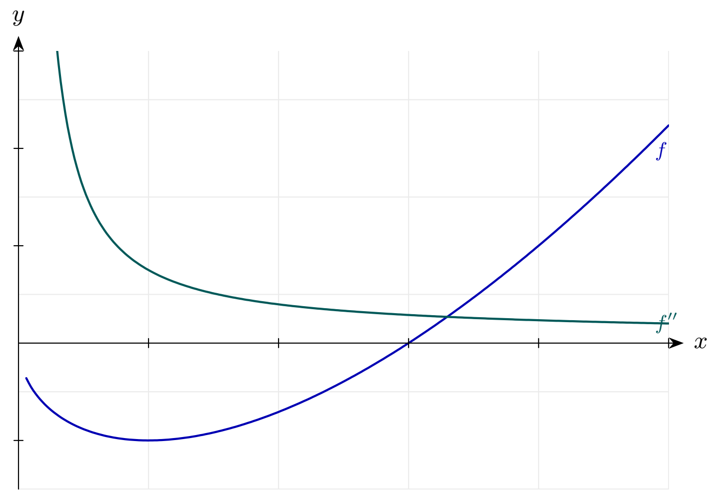
Step 3: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \tfrac{3}{4}x^{-1/2} + \tfrac{3}{4}x^{-3/2} = \frac{3}{4\sqrt{x}} + \frac{3}{4x\sqrt{x}}$$

Differentiate once more and simplify.

Final answer: $\dfrac{d^2y}{dx^2} = \dfrac{3}{4\sqrt{x}} + \dfrac{3}{4x\sqrt{x}}$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 43 `(d3-043)`

Difficulty: Standard | Marks: 3

Given $f(x) = x^4 - 2x^3 + 5x$, find $f''(1)$.

Worked solution.

Step 1: First derivative.

$$f'(x) = 4x^3 - 6x^2 + 5$$

Apply the power rule.

Step 2: Second derivative.

$$f''(x) = 12x^2 - 12x$$

Differentiate once more.

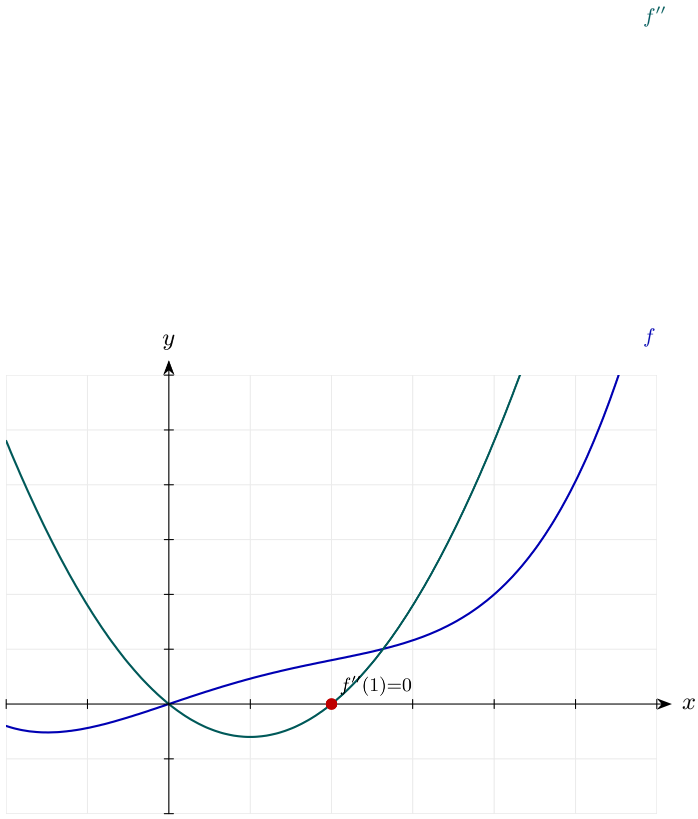
Step 3: Substitute $x = 1$.

$$f''(1) = 12 - 12 = 0$$

Evaluate.

Final answer: $f''(1) = 0$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 44 `(d3-044)`

Difficulty: Standard | Marks: 4

Given $y = 3x^2 - 2\sqrt{x}$, find the value of $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ at $x = 4$.

Worked solution.

Step 1: Rewrite and differentiate.

$$y = 3x^2 - 2x^{1/2} \;\Rightarrow\; \frac{\mathrm{d}y}{\mathrm{d}x} = 6x - x^{-1/2}$$

Apply the power rule.

Step 2: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6 + \tfrac{1}{2}x^{-3/2}$$

Differentiate each term.

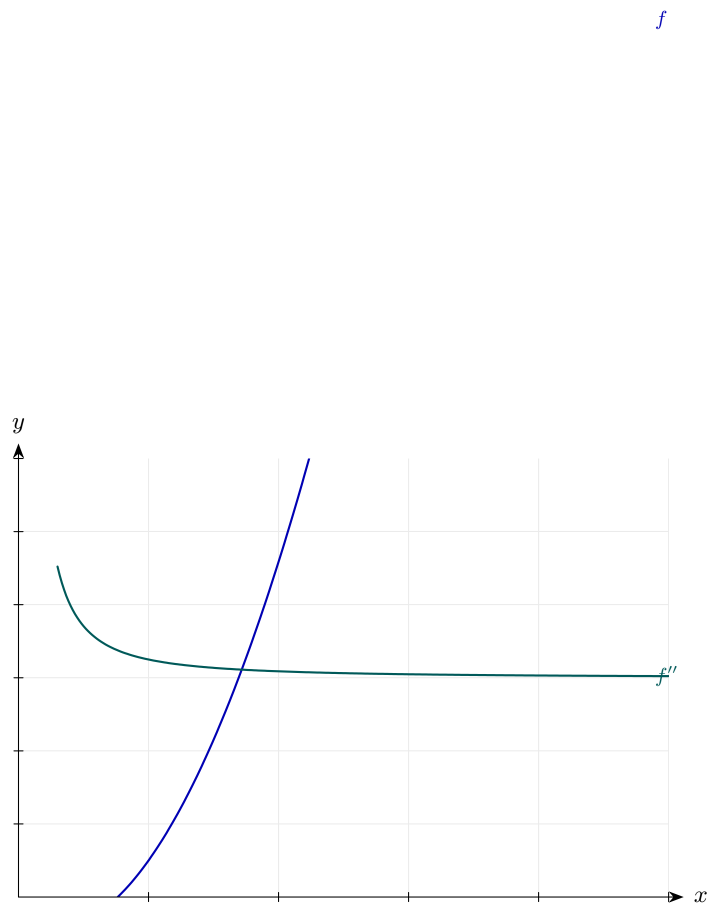
Step 3: Substitute $x = 4$.

$$6 + \tfrac{1}{2}(4)^{-3/2} = 6 + \tfrac{1}{2} \cdot \tfrac{1}{8} = 6 + \tfrac{1}{16} = \tfrac{97}{16}$$

$4^{3/2} = (\sqrt{4})^3 = 8$.

Final answer: $\left.\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=4} = \dfrac{97}{16}$

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 45 `(d3-045)`

Difficulty: Standard | *Marks: 4*

A curve has equation $y = ax^3 + bx^2$. Given that $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12x - 4$, find the values of a and b.

Worked solution.

Step 1: Differentiate twice.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3ax^2 + 2bx \;\Rightarrow\; \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6ax + 2b$$

Apply the power rule.

Step 2: Compare coefficients.

$$6ax + 2b \equiv 12x - 4$$

Match the x term and the constant term.

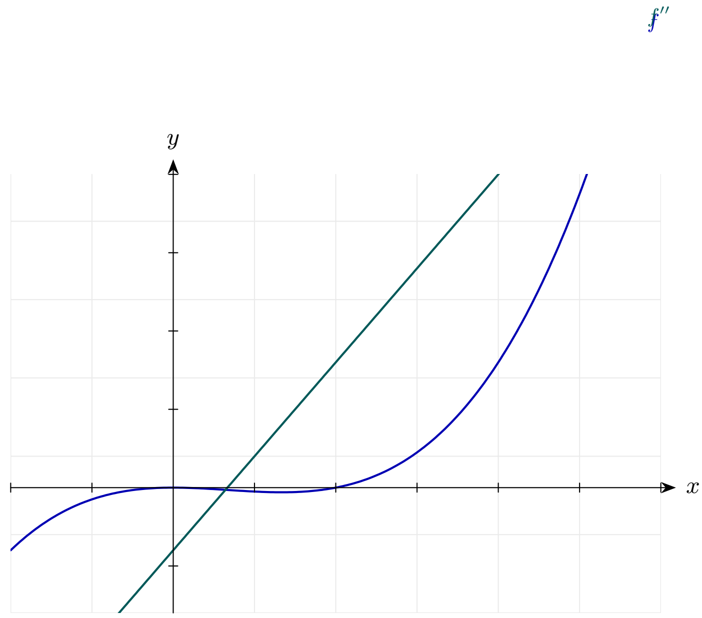
Step 3: Solve.

$$6a = 12 \;\Rightarrow\; a = 2, \quad 2b = -4 \;\Rightarrow\; b = -2$$

Two equations, two unknowns.

Final answer: $a = 2$ and $b = -2$.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 46 `(d3-046)`

Difficulty: Standard | *Marks: 4*

Given $f(x) = x^3 - kx^2 + 3x$, find the value of k for which $f''(2) = 0$.

Worked solution.

Step 1: Differentiate twice.

$$f'(x) = 3x^2 - 2kx + 3, \quad f''(x) = 6x - 2k$$

Apply the power rule.

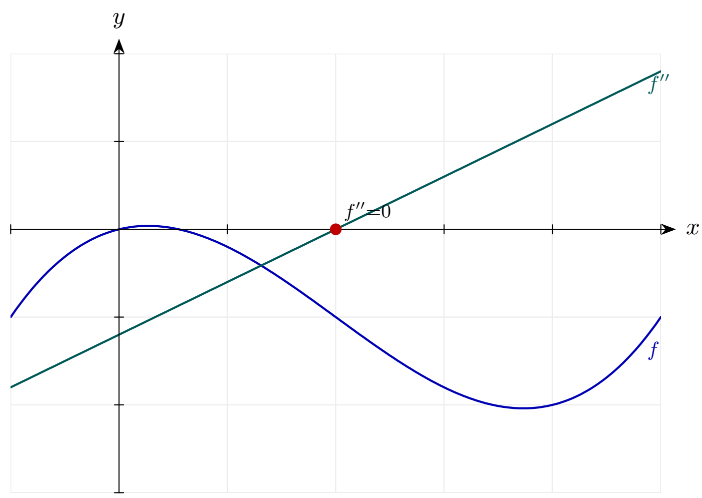
Step 2: Set $f''(2) = 0$.

$$12 - 2k = 0 \Rightarrow k = 6$$

Solve for k.

Final answer: $k = 6$.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 47 `(d3-047)`

Difficulty: Standard | Marks: 3

For $y = 2x^3 + 5x^2 - 4x$, find the value of x at which $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$.

Worked solution.

Step 1: Differentiate twice.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 6x^2 + 10x - 4, \quad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12x + 10$$

Apply the power rule.

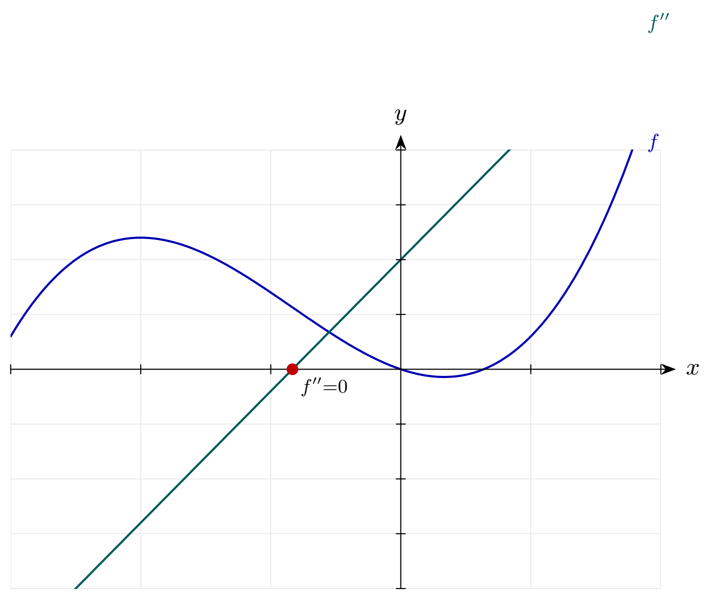
Step 2: Set second derivative to zero.

$$12x + 10 = 0 \ \Rightarrow \ x = -\tfrac{5}{6}$$

Solve for x.

Final answer: $x = -\dfrac{5}{6}$.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 48 `(d3-048)`

Difficulty: Standard | Marks: 4

Given $f(x) = x^4 - 4x^3$, find all values of x for which $f''(x) = 0$.

Worked solution.

Step 1: Differentiate twice.

$$f'(x) = 4x^3 - 12x^2, \quad f''(x) = 12x^2 - 24x$$

Apply the power rule.

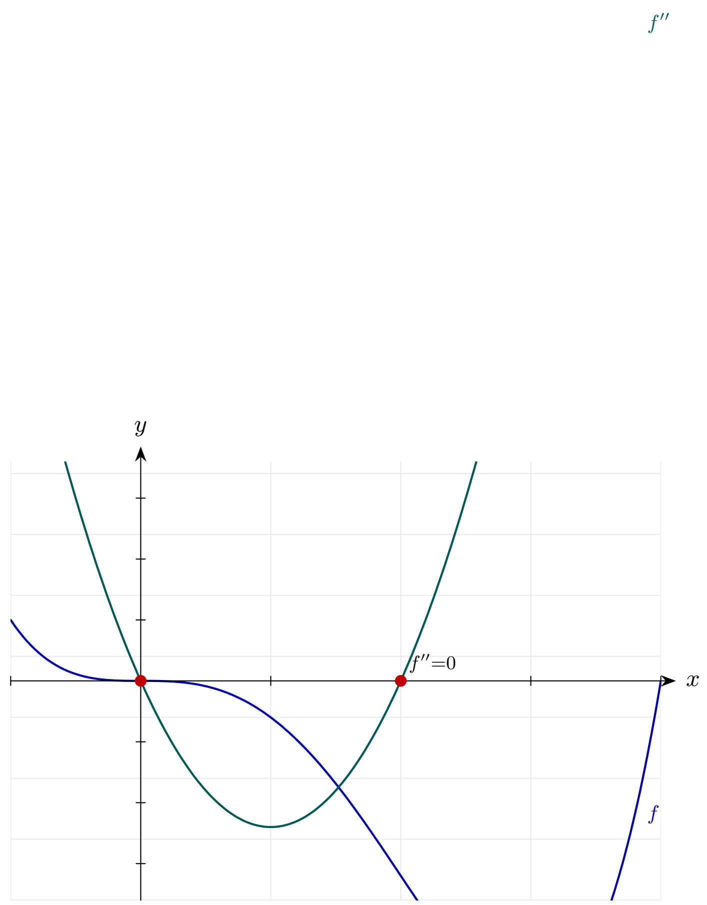
Step 2: Solve $f''(x) = 0$.

$$12x(x-2) = 0 \;\Rightarrow\; x = 0 \text{ or } x = 2$$

Factorise.

Final answer: $x = 0$ or $x = 2$.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 49 (d3-049)

Difficulty: Standard | Marks: 6

Find the stationary points of $y = x^3 - 12x + 5$ and use the second derivative to classify them.

Worked solution.

Step 1: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 12$$

Apply the power rule.

Step 2: Set to zero.

$$3x^2 - 12 = 0 \;\Rightarrow\; x^2 = 4 \;\Rightarrow\; x = \pm 2$$

Solve for stationary points.

Step 3: Find y at each.

$$x = 2 :\; 8 - 24 + 5 = -11$$
$$x = -2 :\; -8 + 24 + 5 = 21$$

Substitute back.

Step 4: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6x$$

Differentiate once more.

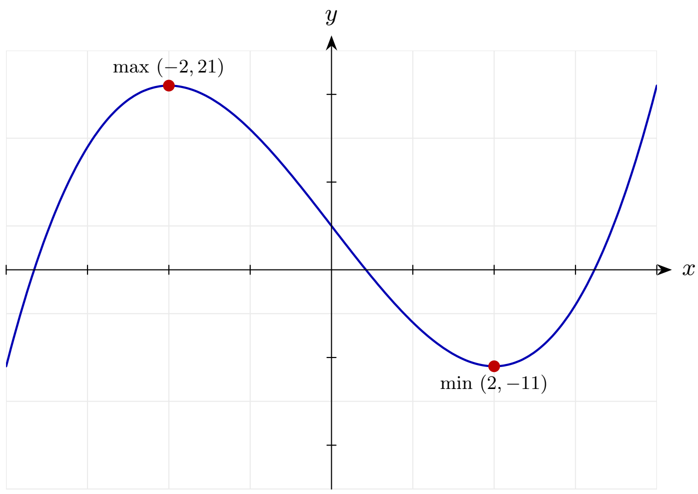
Step 5: Apply the test.

$$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=2} = 12 > 0 \text{ (min)}$$
$$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=-2} = -12 < 0 \text{ (max)}$$

A positive second derivative indicates the curve is concave up (minimum); a negative one indicates concave down (maximum). If it were exactly 0, we would need to examine the gradient on either side instead.

Final answer: Minimum at $(2, -11)$; maximum at $(-2, 21)$.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 50 (d3-050)

Difficulty: Standard / Marks: 7

The curve C has equation $y = 2x^3 + 3x^2 - 36x + 1$. Find its stationary points and classify them using the second derivative.

Worked solution.

Step 1: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 6x^2 + 6x - 36$$

Apply the power rule.

Step 2: Solve for stationary points.

$$6(x^2 + x - 6) = 0 \;\Rightarrow\; (x+3)(x-2) = 0$$

Factorise.

Step 3: Find y at each.

$$x = -3: \;\; -54 + 27 + 108 + 1 = 82$$
$$x = 2: \;\; 16 + 12 - 72 + 1 = -43$$

Substitute.

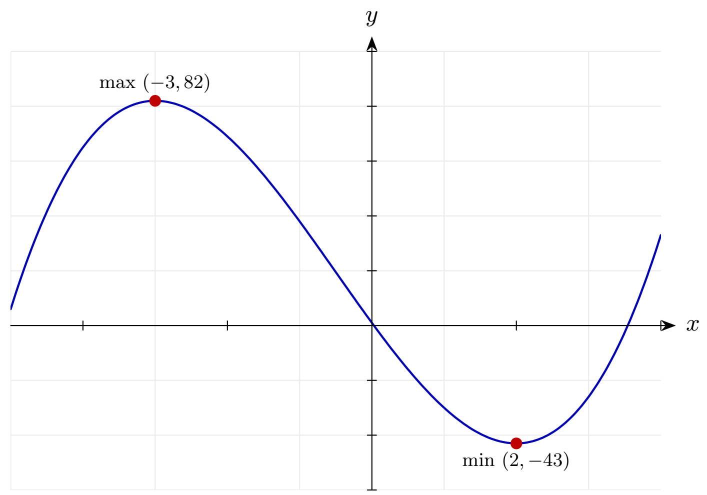
Step 4: Second derivative test.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12x + 6$$
$$\text{At } x = -3: -30 < 0 \text{ (max)}$$
$$\text{At } x = 2: 30 > 0 \text{ (min)}$$

Substitute each x into the second derivative. Negative $\Rightarrow$ concave down (maximum); positive $\Rightarrow$ concave up (minimum).

Final answer: Maximum at $(-3, 82)$; minimum at $(2, -43)$.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 51 (d3-051)

Difficulty: Standard | Marks: 6

Find the stationary points of $f(x) = x^3 - 6x^2 + 9x - 2$ and classify each using $f''(x)$.

Worked solution.

Step 1: Compute $f'(x)$.

$$f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$$

Factorise after taking out 3.

Step 2: Set $f'(x) = 0$.

$$x = 1 \text{ or } x = 3$$

Two stationary points.

Step 3: Find y values.

$$f(1) = 1 - 6 + 9 - 2 = 2$$
$$f(3) = 27 - 54 + 27 - 2 = -2$$

Substitute.

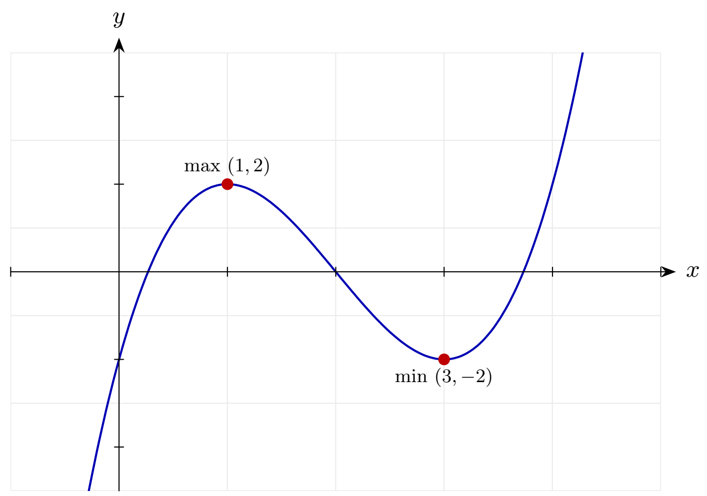
Step 4: Compute $f''(x)$ and test.

$$f''(x) = 6x - 12$$
$$f''(1) = -6 < 0 \text{ (max)}$$
$$f''(3) = 6 > 0 \text{ (min)}$$

Apply the second derivative test: $f''(x) < 0$ at a stationary point means a local maximum, > 0 means a local minimum.

Final answer: Maximum at $(1, 2)$; minimum at $(3, -2)$.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 52 (d3-052)

Difficulty: Standard | Marks: 8

Find the stationary points of $y = x^4 - 4x^3 + 4x^2 + 1$ and classify them.

Worked solution.

Step 1: Differentiate.

$$\frac{dy}{dx} = 4x^3 - 12x^2 + 8x$$

Apply the power rule.

Step 2: Factorise and solve.

$$4x(x^2 - 3x + 2) = 0 \;\Rightarrow\; 4x(x-1)(x-2) = 0$$

Three stationary points: $x = 0, 1, 2$.

Step 3: Find y at each.

$$y(0) = 1, \quad y(1) = 1 - 4 + 4 + 1 = 2, \quad y(2) = 16 - 32 + 16 + 1 = 1$$

Substitute.

Step 4: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12x^2 - 24x + 8$$

Differentiate once more.

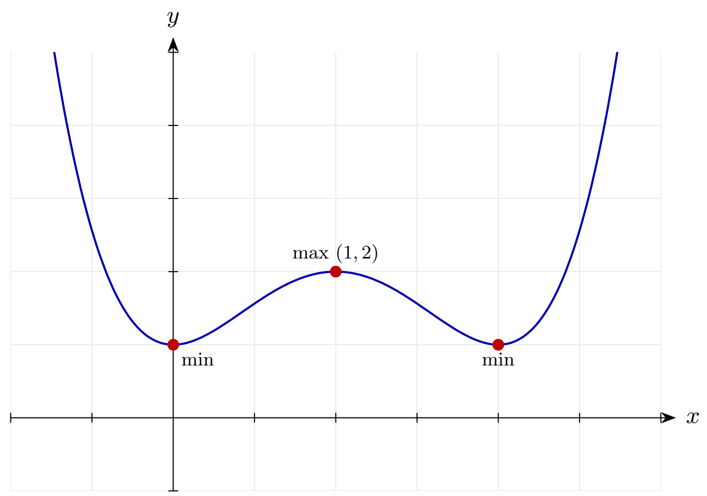
Step 5: Apply the test.

$$x = 0 : 8 > 0 \text{ (min)}$$
$$x = 1 : -4 < 0 \text{ (max)}$$
$$x = 2 : 8 > 0 \text{ (min)}$$

Apply the second derivative test at each stationary point. Positive values give minima (concave up); negative values give a maximum (concave down).

Final answer: Minima at $(0, 1)$ and $(2, 1)$; maximum at $(1, 2)$.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 53 (d3-053)

Difficulty: Standard / Marks: 6

For $y = x + \dfrac{9}{x}$, $x > 0$, find the stationary point and use the second derivative to show that it is a minimum.

Worked solution.

Step 1: Rewrite and differentiate.

$$y = x + 9x^{-1} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = 1 - 9x^{-2}$$

Apply the power rule.

Step 2: Set to zero.

$$1 - \frac{9}{x^2} = 0 \Rightarrow x^2 = 9 \Rightarrow x = 3 \text{ (since } x > 0)$$

Solve for x.

Step 3: Find y.

$$y(3) = 3 + 3 = 6$$

Substitute.

Step 4: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 18x^{-3} = \frac{18}{x^3}$$

Differentiate once more.

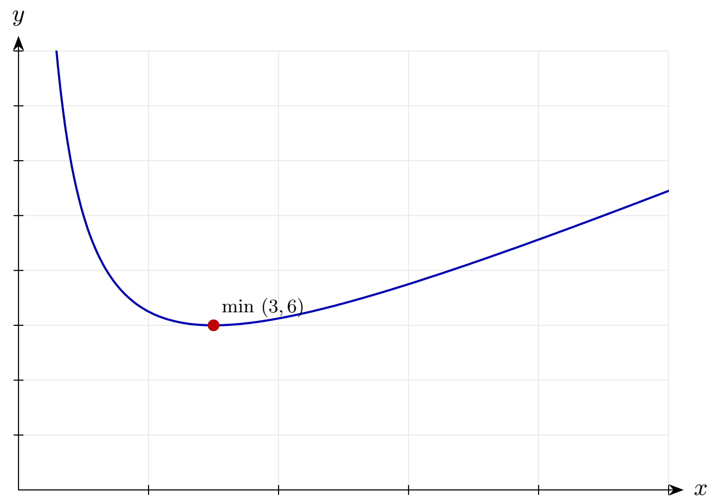
Step 5: Evaluate at $x = 3$.

$$\frac{18}{27} = \frac{2}{3} > 0 \Rightarrow \text{minimum}$$

The second derivative is positive at the stationary point, so the curve is concave up there. This confirms a local minimum.

Final answer: Stationary point at $(3, 6)$; it is a minimum.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 54 (d3-054)

Difficulty: Standard | Marks: 6

The curve C has equation $y = x^3 - 3x$. (a) Find the stationary points. (b) Classify each using the second derivative.

Worked solution.

Step 1: (a) First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 3$$

Apply the power rule.

Step 2: Solve.

$$3x^2 - 3 = 0 \;\Rightarrow\; x^2 = 1 \;\Rightarrow\; x = \pm 1$$

Two stationary points.

Step 3: Find y.

$$y(1) = -2, \quad y(-1) = 2$$

Substitute.

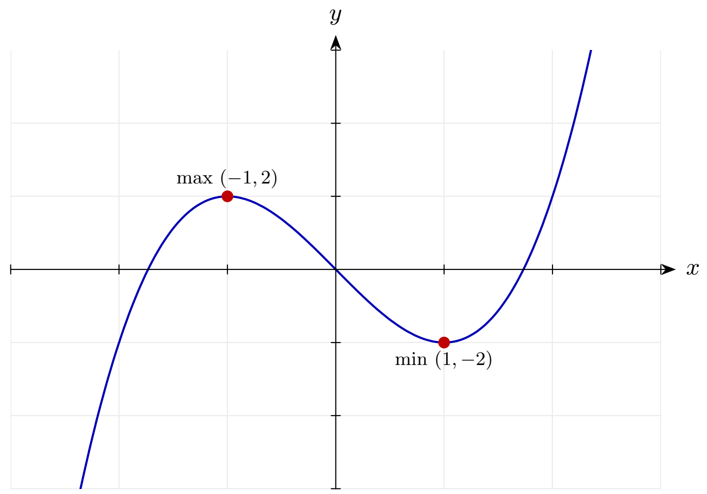
Step 4: (b) Second derivative.

$$\frac{d^2y}{dx^2} = 6x$$
$$x = 1 : 6 > 0 \text{ (min)}$$
$$x = -1 : -6 < 0 \text{ (max)}$$

Substitute each x into the second derivative. Positive value $\Rightarrow$ concave up (minimum); negative value $\Rightarrow$ concave down (maximum).

Final answer: (a) $(1, -2)$ and $(-1, 2)$
(b) $(1, -2)$ minimum, $(-1, 2)$ maximum.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 55 `(d3-055)`

Difficulty: Standard | Marks: 4

Find the range of values of x for which the curve $y = x^3 - 6x^2 + 5$ is concave up.

Worked solution.

Step 1: Differentiate twice.

$$\frac{dy}{dx} = 3x^2 - 12x, \quad \frac{d^2y}{dx^2} = 6x - 12$$

Apply the power rule.

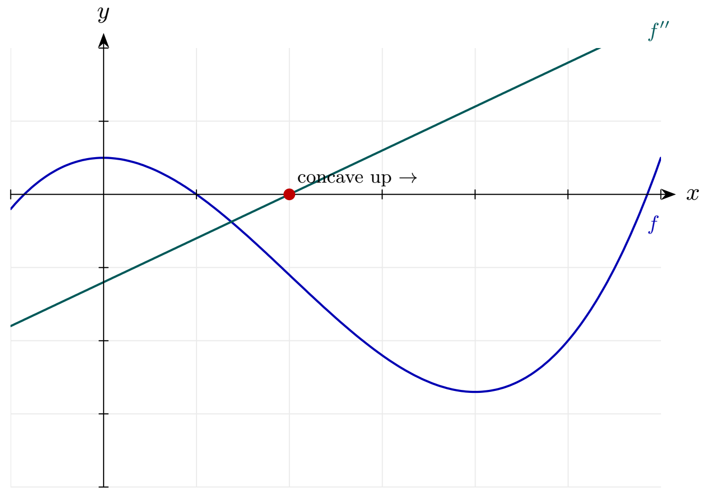
Step 2: Concave up when second derivative is positive.

$$6x - 12 > 0 \;\Rightarrow\; x > 2$$

A curve is concave up where the second derivative is positive — this is where the gradient is increasing. Solve the linear inequality to find the interval.

Final answer: Concave up for $x > 2$.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 56 `(d3-056)`

Difficulty: Standard | Marks: 4

Find the x-coordinate of the point of inflection of $y = x^3 - 9x^2 + 24x - 15$.

Worked solution.

Step 1: Second derivative.

$$\frac{d^2y}{dx^2} = 6x - 18$$

Differentiate twice.

Step 2: Set to zero.

$$6x - 18 = 0 \;\Rightarrow\; x = 3$$

Candidate for inflection.

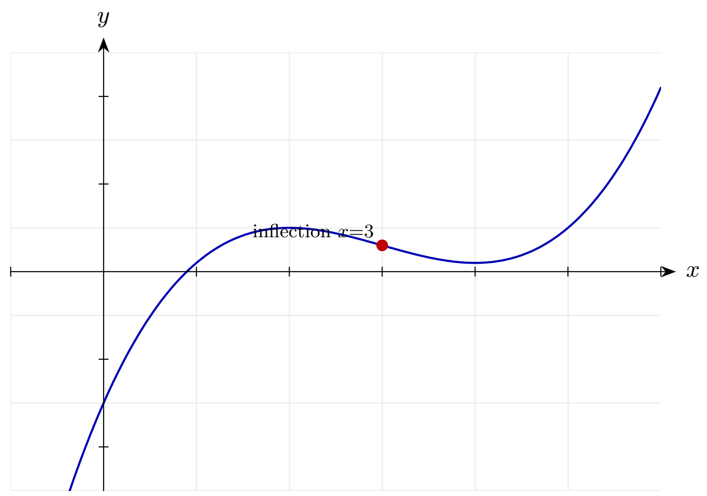
Step 3: Check sign change.

$$f''(2) = -6 < 0, \; f''(4) = 6 > 0$$

For a genuine point of inflection the second derivative must change sign across the candidate x-value. Here it goes from negative to positive, so concavity flips and this is indeed an inflection point.

Final answer: Point of inflection at $x = 3$.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 57 (d3-057)

Difficulty: Standard | Marks: 4

A curve has equation $y = x^4 - 6x^2$. Find the range of values of x for which the curve is concave down.

Worked solution.

Step 1: Differentiate twice.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 4x^3 - 12x, \quad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12x^2 - 12$$

Apply the power rule.

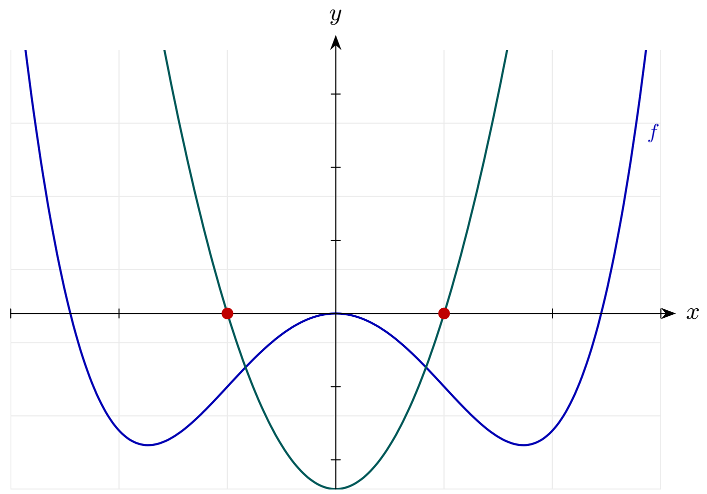
Step 2: Concave down when second derivative is negative.

$$12x^2 - 12 < 0 \;\Rightarrow\; x^2 < 1 \;\Rightarrow\; -1 < x < 1$$

A curve is concave down where its second derivative is negative. Be careful taking the square root of an inequality: $x^2 < 1$ means $|x| < 1$, giving the interval $-1 < x < 1$.

Final answer: Concave down on $-1 < x < 1$.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 58 `(d3-058)`

Difficulty: Standard | Marks: 5

The curve $y = 2x^3 - 9x^2 + 12x$ has a point of inflection. Find its coordinates.

Worked solution.

Step 1: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12x - 18$$

Differentiate twice.

Step 2: Set to zero.

$$12x - 18 = 0 \;\Rightarrow\; x = \tfrac{3}{2}$$

Candidate for inflection.

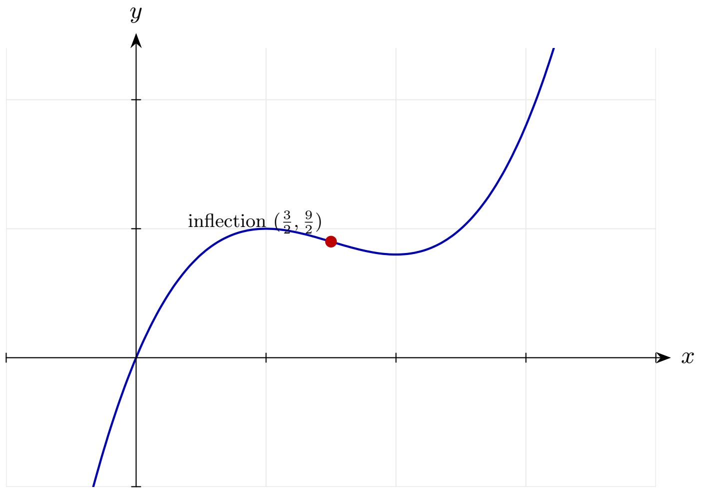
Step 3: Find y.

$$y\left(\tfrac{3}{2}\right) = 2 \cdot \tfrac{27}{8} - 9 \cdot \tfrac{9}{4} + 12 \cdot \tfrac{3}{2} = \tfrac{27}{4} - \tfrac{81}{4} + 18 = -\tfrac{54}{4} + 18 = \tfrac{9}{2}$$

Substitute.

Final answer: Point of inflection at $\left(\tfrac{3}{2}, \tfrac{9}{2}\right)$.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 59 (d3-059)

Difficulty: Standard | Marks: 5

Find the values of x for which the curve $y = x^3 - 3x^2$ is decreasing and concave up.

Worked solution.

Step 1: First derivative.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 6x = 3x(x-2)$$

Factorise.

Step 2: Decreasing when this is negative.

$$3x(x-2) < 0 \;\Rightarrow\; 0 < x < 2$$

Quadratic is negative between roots.

Step 3: Second derivative.

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6x - 6$$

Differentiate again.

Step 4: Concave up when this is positive.

$$6x - 6 > 0 \;\Rightarrow\; x > 1$$

Solve.

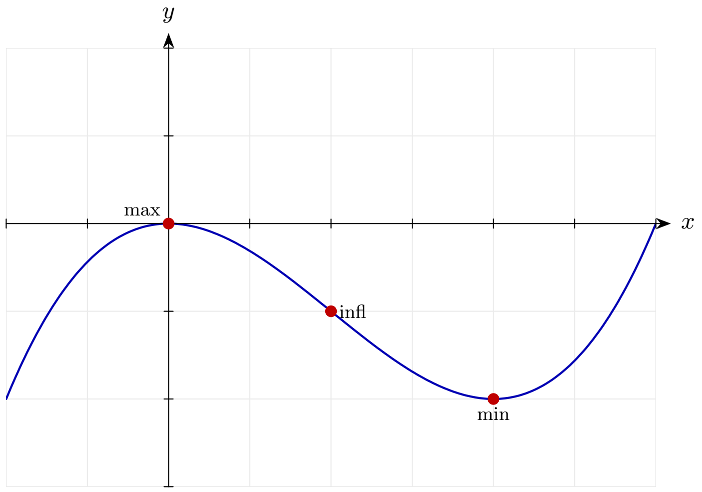
Step 5: Intersect the two conditions.

$$(0 < x < 2) \cap (x > 1) \;\Rightarrow\; 1 < x < 2$$

Both conditions hold here.

Final answer: Decreasing and concave up on $1 < x < 2$.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 60 `(d3-060)`

Difficulty: Standard | Marks: 4

Show that the curve $y = x^4 + 2$ has no point of inflection.

Worked solution.

Step 1: Second derivative.

$$\frac{d^2y}{dx^2} = 12x^2$$

Differentiate twice.

Step 2: Set to zero.

$$12x^2 = 0 \;\Rightarrow\; x = 0$$

Candidate point.

Step 3: Check the sign of the second derivative around $x = 0$.

$$f''(-1) = 12 > 0, \quad f''(1) = 12 > 0$$

The second derivative does not change sign across $x = 0$.

Step 4: Conclusion.

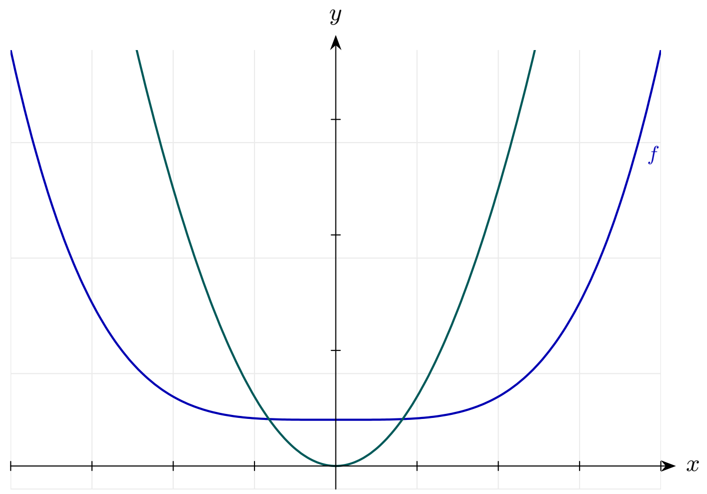
$$\text{No point of inflection at } x = 0 \checkmark$$

A point of inflection requires $\frac{d^2y}{dx^2}$ to change sign across the point. Here $12x^2 \geq 0$ for all x, so it never changes sign — there is no inflection point anywhere on the curve.

Final answer: No point of inflection — $f''(x) = 12x^2 \geq 0$ everywhere.

Diagram.

f''



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 61 `(d3-061)`

Difficulty: Standard | Marks: 6

A rectangular enclosure has perimeter 20 m. Its area A (in m^2) is given by $A = x(10-x)$, where x is the length of one side. Find the value of x that maximises A, and use the second derivative to justify that it is a maximum.

Worked solution.

Step 1: Expand.

$$A = 10x - x^2$$

Remove the brackets.

Step 2: Differentiate.

$$\frac{\mathrm{d}A}{\mathrm{d}x} = 10 - 2x$$

Apply the power rule.

Step 3: Set to zero.

$$10 - 2x = 0 \Rightarrow x = 5$$

Solve.

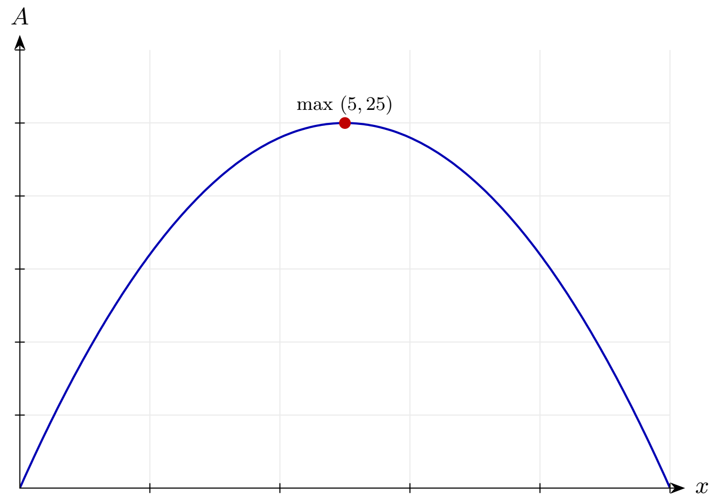
Step 4: Second derivative test.

$$\frac{\mathrm{d}^2A}{\mathrm{d}x^2} = -2 < 0 \Rightarrow \text{maximum}$$

The second derivative is the constant -2, so the area function is concave down everywhere. This confirms the stationary point is a maximum.

Final answer: $x = 5$ m gives the maximum area (a square of side 5 m).

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 62 (d3-062)

Difficulty: Standard | Marks: 6

The height of a projectile h metres after t seconds is $h = 25t - 5t^2$. (a) Find $\frac{dh}{dt}$ and $\frac{d^2h}{dt^2}$. (b) Find the time at which the height is maximum and state the maximum height. (c) Interpret the sign of $\frac{d^2h}{dt^2}$.

Worked solution.

Step 1: (a) Differentiate twice.

$$\frac{dh}{dt} = 25 - 10t, \quad \frac{d^2h}{dt^2} = -10$$

Apply the power rule.

Step 2: (b) Maximum when velocity is zero.

$$25 - 10t = 0 \Rightarrow t = 2.5$$

Solve.

Step 3: Maximum height.

$$h(2.5) = 62.5 - 31.25 = 31.25 \text{ m}$$

Substitute.

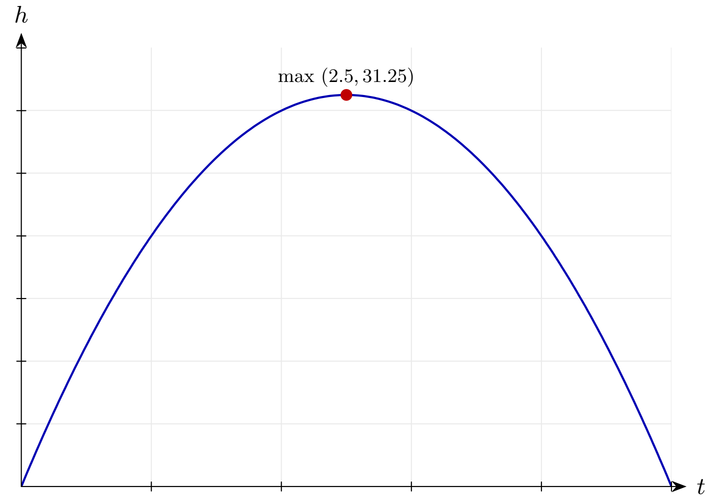
Step 4: (c) Interpretation.

$$\frac{d^2h}{dt^2} = -10 \text{ m/s}^2$$

The second derivative of position with respect to time is acceleration. A constant -10 m/s^2 is gravity acting downwards, and its negative sign also tells us the height function is concave down — consistent with the projectile turning around at a maximum.

Final answer: (a) $\frac{dh}{dt} = 25 - 10t$, $\frac{d^2h}{dt^2} = -10$
(b) $t = 2.5$ s, max height 31.25 m
(c) Acceleration is -10 m/s^2 (downward).

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 63 (d3-063)

Difficulty: Standard | *Marks: 8*

An open-top box is made from a square sheet of side 12 cm by cutting a square of side x cm from each corner and folding up the sides. The volume is $V = x(12 - 2x)^2$ cm^3. Find the value of x that maximises the volume and verify it is a maximum using the second derivative.

Worked solution.

Step 1: Expand.

$$V = x(144 - 48x + 4x^2) = 4x^3 - 48x^2 + 144x$$

Multiply out.

Step 2: Differentiate.

$$\frac{dV}{dx} = 12x^2 - 96x + 144$$

Apply the power rule.

Step 3: Set to zero.

$$12(x^2 - 8x + 12) = 0 \Rightarrow (x - 2)(x - 6) = 0$$

Factorise.

Step 4: Valid solution.

$$x = 2 \text{ (since } 0 < x < 6)$$

For a real box, $12 - 2x > 0$, so $x < 6$.

Step 5: Second derivative test.

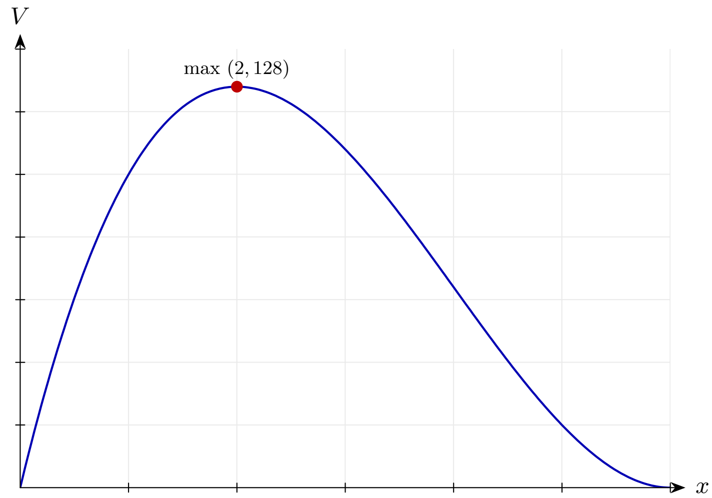
$$\frac{\mathrm{d}^2V}{\mathrm{d}x^2} = 24x - 96$$

$$\left.\frac{\mathrm{d}^2V}{\mathrm{d}x^2}\right|_{x=2} = 48 - 96 = -48 < 0$$

A negative second derivative at the stationary point means the volume function is concave down there, confirming $x = 2$ gives a maximum (not a minimum).

Final answer: $x = 2$ cm gives the maximum volume; $V(2) = 128$ cm^3.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 64 (d3-064)

Difficulty: Standard | Marks: 5

The population P (in thousands) of a species t years after a study begins is modelled by $P = 20 + 6t - t^2$. (a) Find the time at which the population is greatest and its value. (b) State the sign of $\frac{\mathrm{d}^2P}{\mathrm{d}t^2}$ and interpret it.

Worked solution.

Step 1: (a) Differentiate.

$$\frac{\mathrm{d}P}{\mathrm{d}t} = 6 - 2t$$

Rate of change of population.

Step 2: Set to zero.

$$6 - 2t = 0 \;\Rightarrow\; t = 3$$

Critical time.

Step 3: Maximum population.

$$P(3) = 20 + 18 - 9 = 29$$

Substitute.

Step 4: (b) Second derivative.

$$\frac{\mathrm{d}^2 P}{\mathrm{d}t^2} = -2 < 0$$

The second derivative is a constant -2, negative throughout. This means the rate of change of P (the growth rate) is itself always decreasing.

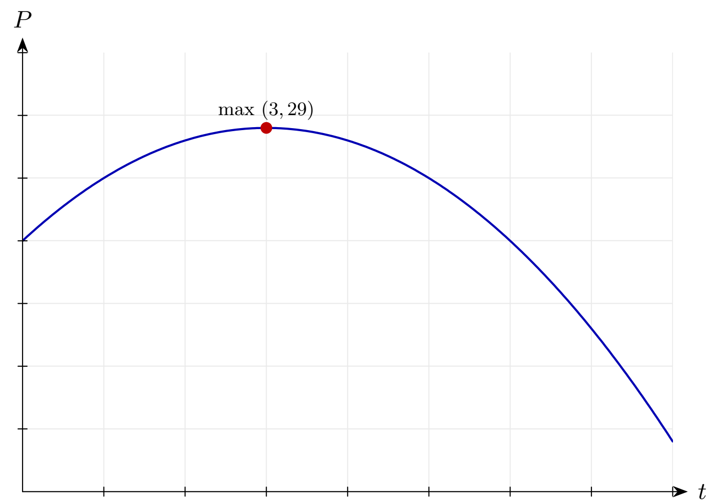
Step 5: Interpretation.

$$\text{Rate of growth is always decreasing.}$$

The population grows more slowly over time, then declines.

Final answer: (a) Maximum population is 29 thousand at $t = 3$ years.
(b) $\frac{\mathrm{d}^2 P}{\mathrm{d}t^2} = -2$, the growth rate is always decreasing.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 65 (d3-065)

Difficulty: Standard | Marks: 7

A cylinder has surface area $S = 2\pi r^2 + \dfrac{200}{r}$ for $r > 0$. Find the value of r that minimises S, and use the second derivative to confirm it is a minimum.

Worked solution.

Step 1: Rewrite and differentiate.

$$S = 2\pi r^2 + 200r^{-1} \Rightarrow \frac{\mathrm{d}S}{\mathrm{d}r} = 4\pi r - 200r^{-2}$$

Apply the power rule.

Step 2: Set to zero.

$$4\pi r = \frac{200}{r^2} \Rightarrow r^3 = \frac{50}{\pi} \Rightarrow r = \sqrt[3]{\frac{50}{\pi}}$$

Rearrange.

Step 3: Second derivative.

$$\frac{\mathrm{d}^2S}{\mathrm{d}r^2} = 4\pi + 400r^{-3} = 4\pi + \frac{400}{r^3}$$

Differentiate once more.

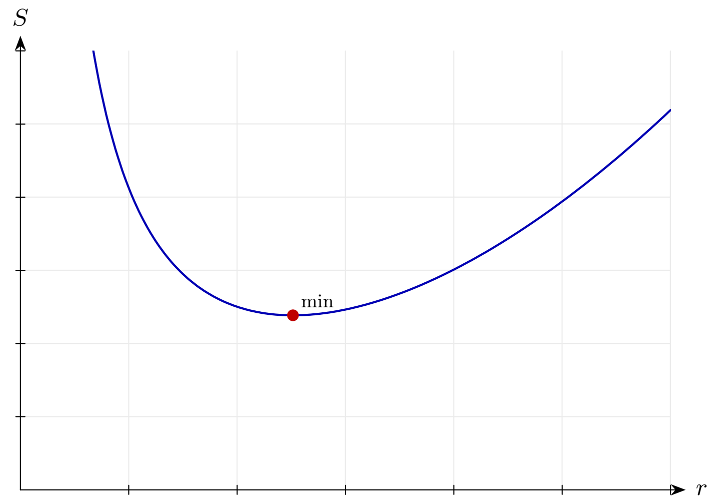
Step 4: Confirm minimum.

$$r > 0 \;\Rightarrow\; \frac{\mathrm{d}^2S}{\mathrm{d}r^2} > 0$$

Both 4π and $\frac{400}{r^3}$ are positive when $r > 0$, so the second derivative is strictly positive on the domain. Concave up at the stationary point means a local minimum.

Final answer: $r = \sqrt[3]{\frac{50}{\pi}}$ gives the minimum surface area.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 66 (d3-066)

Difficulty: Challenge | Marks: 8

The curve C has equation $y = x^3 - 3ax^2 + 3a^2x$, where a is a positive constant. (a) Find the stationary points of C in terms of a. (b) Find the nature of each stationary point using the second derivative.

Worked solution.

Step 1: (a) Differentiate.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 6ax + 3a^2$$

Apply the power rule.

Step 2: Set to zero and simplify.

$$3(x^2 - 2ax + a^2) = 0 \;\Rightarrow\; (x-a)^2 = 0 \;\Rightarrow\; x = a$$

Repeated root.

Step 3: Find y.

$$y(a) = a^3 - 3a \cdot a^2 + 3a^2 \cdot a = a^3$$

Substitute.

Step 4: (b) Second derivative.

$$\frac{d^2y}{dx^2} = 6x - 6a$$

Differentiate once more.

Step 5: Evaluate at $x = a$.

$$6a - 6a = 0$$

Second derivative test is inconclusive.

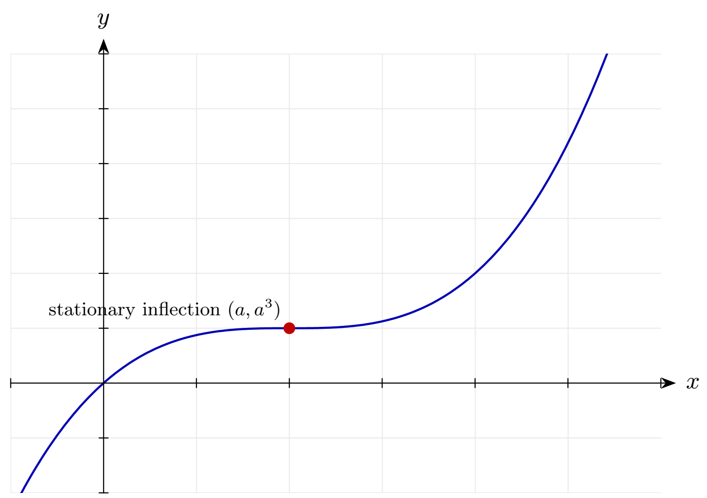
Step 6: Check sign change around $x = a$.

$$f'(a - \epsilon) > 0, \quad f'(a + \epsilon) > 0$$

When $\frac{d^2y}{dx^2} = 0$ the second-derivative test is inconclusive. Falling back on first-derivative behaviour: $\frac{dy}{dx} = 3(x - a)^2 \geq 0$ for all x, so the gradient does not change sign — this is a stationary point of inflection, not a turning point.

Final answer: (a) One stationary point at (a, a^3).
(b) It is a stationary point of inflection.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 67 (d3-067)

Difficulty: Challenge | Marks: 8

A function is defined by $f(x) = x^3 + px^2 + qx + 1$, where p and q are constants. The curve $y = f(x)$ has a stationary point at $(2, -3)$. (a) Find p and q. (b) Determine the nature of the stationary point.

Worked solution.

Step 1: (a) Use $f(2) = -3$.

$$8 + 4p + 2q + 1 = -3 \;\Rightarrow\; 4p + 2q = -12 \;\Rightarrow\; 2p + q = -6$$

First equation.

Step 2: Use $f'(2) = 0$.

$$f'(x) = 3x^2 + 2px + q \;\Rightarrow\; f'(2) = 12 + 4p + q = 0$$

Stationary point condition.

Step 3: Solve the simultaneous equations.

$$\begin{cases} 2p + q = -6 \\ 4p + q = -12 \end{cases} \quad\Rightarrow\; 2p = -6 \;\Rightarrow\; p = -3,\; q = 0$$

Subtract the first from the second.

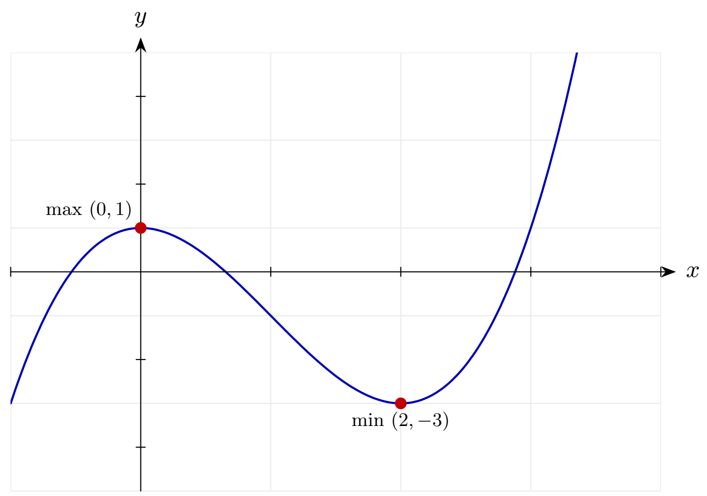
Step 4: (b) Second derivative.

$$f''(x) = 6x + 2p = 6x - 6$$
$$f''(2) = 12 - 6 = 6 > 0$$

Substitute $p = -3$ into $f''(x) = 6x + 2p$, then evaluate at $x = 2$. A positive value means concave up — a local minimum.

Final answer: (a) $p = -3,\; q = 0$.
(b) The stationary point is a minimum.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 68 (d3-068)

Difficulty: Challenge | Marks: 10

The curve C has equation $y = x^4 - 4x^3 + 4x^2$. (a) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. (b) Find the stationary points and classify them. (c) Find the coordinates of any points of inflection.

Worked solution.

Step 1: (a) Differentiate twice.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 4x^3 - 12x^2 + 8x, \quad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12x^2 - 24x + 8$$

Apply the power rule.

Step 2: (b) Factorise and solve $y' = 0$.

$$4x(x^2 - 3x + 2) = 0 \;\Rightarrow\; 4x(x-1)(x-2) = 0$$

Three solutions: $x = 0, 1, 2$.

Step 3: Find y values.

$$y(0) = 0, \quad y(1) = 1 - 4 + 4 = 1, \quad y(2) = 16 - 32 + 16 = 0$$

Substitute.

Step 4: Apply the second derivative test.

$$y''(0) = 8 > 0 \text{ (min)}$$
$$y''(1) = 12 - 24 + 8 = -4 < 0 \text{ (max)}$$
$$y''(2) = 48 - 48 + 8 = 8 > 0 \text{ (min)}$$

Substitute each stationary point into the second derivative. Positive values are local minima (concave up); the negative value at $x = 1$ is a local maximum (concave down).

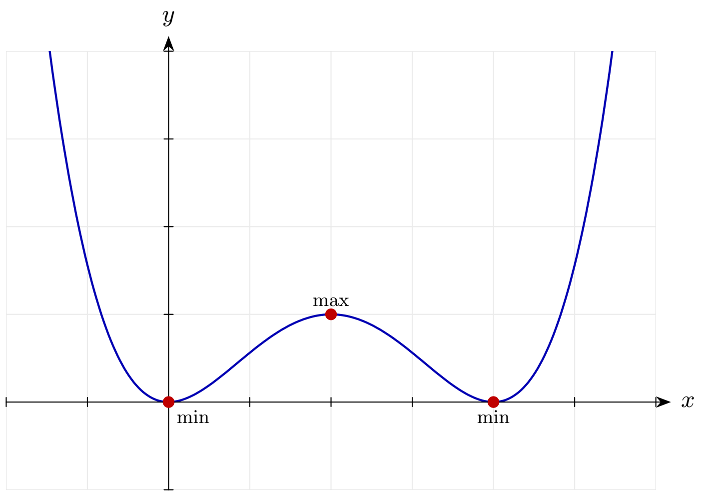
Step 5: (c) Solve $y'' = 0$.

$$12x^2 - 24x + 8 = 0 \;\Rightarrow\; 3x^2 - 6x + 2 = 0 \;\Rightarrow\; x = \frac{6 \pm \sqrt{12}}{6} = 1 \pm \frac{\sqrt{3}}{3}$$

Use the quadratic formula.

Final answer: (a) $y' = 4x^3 - 12x^2 + 8x$, $y'' = 12x^2 - 24x + 8$
(b) Minima at $(0, 0)$ and $(2, 0)$; maximum at $(1, 1)$
(c) Inflection at $x = 1 \pm \frac{\sqrt{3}}{3}$.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 69 (d3-069)

Difficulty: Challenge | Marks: 9

An open rectangular tank has a square base of side x metres and height h metres. Its volume is $32\ \mathrm{m}^3$. (a) Show that its external surface area is $S = x^2 + \dfrac{128}{x}$. (b) Find the value of x that minimises S, using the second derivative to confirm it is a minimum.

Worked solution.

Step 1: (a) Volume condition.

$$x^2 h = 32 \;\Rightarrow\; h = \frac{32}{x^2}$$

Solve for h.

Step 2: Surface area (base + four sides, open top).

$$S = x^2 + 4xh = x^2 + 4x \cdot \frac{32}{x^2} = x^2 + \frac{128}{x} \checkmark$$

As required.

Step 3: (b) Differentiate.

$$S = x^2 + 128x^{-1} \;\Rightarrow\; \frac{\mathrm{d}S}{\mathrm{d}x} = 2x - 128x^{-2}$$

Apply the power rule.

Step 4: Set to zero.

$$2x = \frac{128}{x^2} \;\Rightarrow\; x^3 = 64 \;\Rightarrow\; x = 4$$

Solve.

Step 5: Second derivative.

$$\frac{\mathrm{d}^2S}{\mathrm{d}x^2} = 2 + 256x^{-3}$$

Differentiate once more.

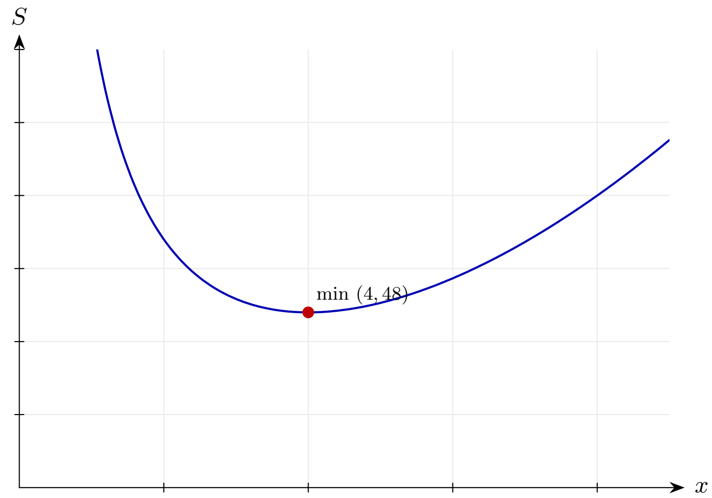
Step 6: Evaluate at $x = 4$.

$$2 + \frac{256}{64} = 2 + 4 = 6 > 0 \;\Rightarrow\; \text{minimum}$$

The second derivative is positive at $x = 4$, so the surface area function is concave up there. This confirms a local minimum.

Final answer: (a) Shown.
(b) $x = 4$ m gives the minimum surface area.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.

Second Order Derivatives 70 `(d3-070)`

Difficulty: Challenge | *Marks: 10*

A function is defined by $f(x) = 2x^3 - 15x^2 + 24x + 7$. (a) Find $f'(x)$ and $f''(x)$. (b) Find the coordinates of the stationary points. (c) Use the second derivative test to determine the nature of each. (d) State the range of values of x for which $f(x)$ is concave up.

Worked solution.

Step 1: (a) Differentiate.

$$f'(x) = 6x^2 - 30x + 24, \quad f''(x) = 12x - 30$$

Apply the power rule.

Step 2: (b) Solve $f'(x) = 0$.

$$6(x^2 - 5x + 4) = 0 \Rightarrow (x - 1)(x - 4) = 0$$

Factorise.

Step 3: Find y values.

$$f(1) = 2 - 15 + 24 + 7 = 18$$
$$f(4) = 128 - 240 + 96 + 7 = -9$$

Substitute.

Step 4: (c) Second derivative test.

$$f''(1) = 12 - 30 = -18 < 0 \text{ (max)}$$
$$f''(4) = 48 - 30 = 18 > 0 \text{ (min)}$$

Substitute each stationary point into $f''(x)$. Negative $\Rightarrow$ concave down (local maximum); positive $\Rightarrow$ concave up (local minimum).

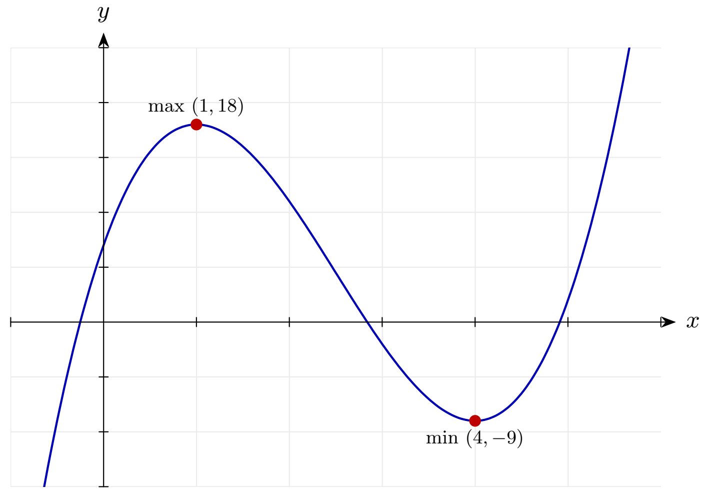
Step 5: (d) Concave up where $f''(x) > 0$.

$$12x - 30 > 0 \Rightarrow x > \tfrac{5}{2}$$

A curve is concave up exactly where its second derivative is positive. Solve the linear inequality directly — no sign flip needed.

Final answer: (a) $f'(x) = 6x^2 - 30x + 24$, $f''(x) = 12x - 30$
(b) $(1, 18)$ and $(4, -9)$
(c) Maximum at $(1, 18)$; minimum at $(4, -9)$
(d) Concave up for $x > \frac{5}{2}$.

Diagram.



Curve (blue); second derivative f'' (teal) or classified stationary points as relevant.